

These findings confirm that well-designed TL strategies—especially FTF and ZS—can reliably compensate for limited data in heating load forecasting. Future work should explore adaptive freezing schemes, learning rate adaptation, and the transferability of the approach to longer forecast horizons.

Nomenclature

Acronyms

ARIMA	Autoregressive Integrated Moving Average
CV-RMSE	Coefficient of Variation of RMSE
FH	Forecast Hour
FHor	Forecast Horizon
FI	Feature Importance
FT	Fine-Tuning
FTF	Fine-Tuning with Frozen Layers
LSTM	Long Short-Term Memory
MAPE	Mean Absolute Percentage Error
MLP	Multilayer Perceptron
NTL	No Transfer Learning
PatchTST	Patch-based Time Series Transformer
PFI	Permutation Feature Importance
SHAP	SHapley Additive exPlanations
SMAPE	Symmetric Mean Absolute Percentage Error
TFT	Temporal Fusion Transformer
TiDE	Time-series Dense Encoder
Time-LLM	Time Large Language Model
TimeMixer	Token-Mixing Time Series Model
TL	Transfer Learning
XAI	Explainable Artificial Intelligence

Latin Symbols

ACF_{24h}	Autocorrelation at 24-hour lag (-)
r	Pearson correlation coefficient (-)
R^2	Coefficient of determination (-)
R_{DN}	Day-Night load ratio (-)
R_{WDWE}	Weekday-Weekend load ratio (-)
$S_{>15^\circ C}$	Share of load above 15 °C (-)
MAE	Mean Absolute Error (-)
NRMSE	Normalised Root Mean Square Error (-)

Subscripts

IG	Internal gains
GHI	Global Horizontal Irradiance
out	Outdoor
t	Time index

1. Introduction

Between 2014 and 2024, the increase in global electricity demand for space and water heating exceeded that for electric transport. Heating will remain an essential factor in demand growth in the coming years. The unlocked potential for load flexibility opens up promising opportunities in the heating sector. However, the consumption patterns of buildings are significantly decoupled from renewable energy generation, particularly during the early evening. During cold or hot spells, the additional demand for heating and cooling can therefore pose a significant threat to grid stability. Based on current digitisation trends, the use of AI could save around 5 % of total electricity consumption for heating and cooling by 2035. In addition, AI-enabled load flexibility in the heating sector alone enables a shift of more than 100 GW in

advanced economies, representing more than 5 % of average peak load [1].

Accurate heating load forecasting plays a central role in this process, enabling the use of demand-side flexibility, the optimisation of heating systems such as heat pumps, and the integration of intermittent renewable energy sources [2]. Data-driven approaches for time series forecasting, particularly machine learning (ML) models, are becoming increasingly important because they can capture complex relationships in building energy data without the need for extensive physical modelling [3,4]. In addition, explainable artificial intelligence (XAI) methods such as Shapley additive explanations (SHAP) help to better understand model decisions and support trust in ML-based forecasts.

However, a key challenge for these data-driven models is that they require large amounts of historical data to be effective [5]. In practice, many buildings, particularly new or recently renovated ones, suffer from data scarcity, which can result in poor model performance and overfitting [6,7]. Transfer learning (TL) has emerged as a promising solution to this problem. It uses knowledge learned from data-rich source domains (e.g. similar buildings with sufficient data) to improve predictions in data-poor target domains [8]. Common TL strategies for building energy prediction often involve propagating model parameters or fine-tuning pre-trained models [7,9,10].

While TL approaches typically focus on adjusting pre-trained models or weights, the use of feature importance (FI) values as static features (SF) has not been explored to date. FI depends heavily on the building type and usage context, and contains valuable implicitly learned information about the building. [11]. Rather than using physically informed neural networks (PINN) [12,13]—which are data-driven models augmented with physics constraints—we adopt an unconstrained data-driven approach. In practice, heating loads also depend on occupancy and control policies, both of which are difficult to observe. Given the scale and diversity of our pre-training corpus, we hypothesise that data-centric transfer may already capture huge parts of the physical and operational regularities.

This study investigates a novel TL-based approach for short-term heating load forecasting that makes targeted use of FI. In this approach, SHAP-based FI values and load-pattern metrics derived from a data-rich source environment are employed as static exogenous features for the forecasting model in the target building. The core idea is to transfer not only model parameters, but also explicit knowledge about characteristic load-pattern metrics and the relative importance of input features—quantified through FI values. This should enable more effective model fitting, especially under conditions of limited data availability. Thus, the approach bridges the gap between purely data-driven black-box methods and the underlying building characteristics implicitly contained in the FI metrics. It is systematically evaluated using 116 datasets and seven time series architectures, ranging from commonly used baselines to state-of-the-art models, namely multilayer perceptron (MLP), long short-term memory (LSTM), temporal fusion transformer (TFT), patch-based time series transformer (PatchTST), time-series dense encoder (TiDE), token-mixing time series model (TimeMixer) and time large language model (Time-LLM).

1.1. Related work

TL plays an increasingly important role in heating load forecasting to address data scarcity by transferring knowledge from data-rich source domains to data-constrained target domains [14,15]. This section provides an overview of the main TL strategies, emphasises the potential benefits of SF, and explores advanced modelling architectures that are relevant to this study. Table 1 provides a summary of the studies.

Table 1

Selected literature on TL for load forecasting, highlighting TL strategies, fine-tuning aspects, and handling of features. The results are indicative and specific to the study conditions.

Work	Domain focus	Datasets	Forecast horizon	TL strategy highlight/fine-tuning	Models	Key reported result (Metric & Improv.)
Gao et al. [16]	Office bldg. energy	3 bldgs (1S, 2T)	24 h	Parameter (fine-tuning last layers)	Seq2Seq, CNN	19.7/20.5 percent points MAPE improv.
Li et al. [17]	Cross-bldg. energy	BDG2 (16 bldgs)	Daily	Param. (all layers fine-tuned), Feat. (DANN)	LSTM, DTLs	+0.75 PIR (DANN)
Park et al. [7]	Cross-bldg. energy	Public dataset	1 h	Param. (stacking, fine-tune last FC layer)	CNN, LSTM, GRU	MAE reduc. 6.4% to 26.36%
Lu et al. [18]	Bldg. heating load	Multi-source (500S, 50T), limited target	24 h	Param. (Meta-learning for model initialisation)	LSTM-based	Up to 61.6% error reduc.
Spencer et al. [14]	Bldg. energy	BDG2	24 h/48 h	Param. (6 TL strategies, incl. fine-tuning of multi-source pre-trained models)	Trf, Inf, PatchTST	PatchTST fine-tune: −15.9% MAE avg.
López Santos et al. [15]	Local energy comm.	Real + synthetic	ST (24 h)	Param. (fine-tuning pre-trained TFT for new users)	TFT	CV-RMSE & SMAPE>40% reduc.
Ahn and Kim [19]	Bldg. power	Sim. (source) + Real (target, 24 h)	Hourly	Param. (pre-train sim, fine-tune real with 24 h data)	LSTM	MAPE<10% with 24 h target data
Liao et al. [20]	Load forecast (general)	5 real load datasets (various origins)	1 h–24 h	Param. (fine-tuning pre-trained TimeGPT on scarce load data)	TimeGPT	Outperforms benchmarks in data-scarce (3–7 days of data) for ST horizons; advantage diminishes with more data.
Mohammadi et al. [21]	Household energy consumption	Smart Meters London (1.6M rec.)	ST (daily)	Exogenous variable incorp.; Model-specific hyperparam. tuning (Optuna)	LSTM, DeepAR, TFT, Time-LLM	TFT (+exog, 500k data): 18.8% sMAPE. Permutation FI >TFT attention for global FI. LSTM/Time-LLM is less effective with exog.

Parameter-based transfer learning and model architectures

Parameter-based TL, primarily by fine-tuning pre-trained models, dominates energy forecast for buildings [22]. Gao et al. [16] showed that transfer learning with fine-tuning of sequence-to-sequence (seq2seq) and convolutional neural network (CNN) models improved mean absolute percentage error (MAPE) by 19.7 and 20.5 percentage points, respectively, compared to a baseline LSTM. Similarly, Park et al. [7] achieved a mean absolute error (MAE) reduction of 6.4% to 26.36% by fine-tuning only the last fully connected layers in a stacked ensemble of CNN, LSTM, and gated recurrent units (GRU) models, highlighting the effectiveness of minimal parameter updates. Ahn and Kim [19] showed that LSTMs pre-trained on simulated reference building data could achieve an MAPE below 10% on real target buildings with only 24 h of fine-tuning data. Multi-source TL, as explored by Peng et al. [6], uses an ensemble of LSTMs fine-tuned on multiple selected source domains and outperforms single-source methods by up to 20% to 30%. Meta-learning approaches further improve fast adaptation; Lu et al. [18] reported error reductions of up to 61.6% by using meta-learned LSTM initialisations.

Several studies have demonstrated the potential of transformer architectures. Spencer et al. [14] evaluated six TL strategies for the vanilla Transformer, Informer, and PatchTST models on the building data genome 2 (BDG2) dataset. They found that fine-tuning a PatchTST model pre-trained on multiple source datasets yielded an average MAE reduction of 15.9%. Their findings highlight the importance of selecting an appropriate TL strategy (e.g. fine-tuning on multiple sources versus zero-shot transfer) and ensuring compatibility of feature spaces.

The TFT architecture is specifically designed for time series forecasting. Lim et al. [23] developed the TFT architecture, combining temporal attention with static covariate encoders for processing SF, thereby enhancing multi-horizon forecasting by incorporating contextual information. Without static encoders, the error in the 90th percentile increased by 2.6%. López Santos et al. [15] pre-trained TFTs on data from local energy community buildings, adapting them for new, data-poor users and achieving a reduction of over 40% in coefficient of variation of root mean square error (CV-RMSE) and symmetric mean

absolute percentage error (SMAPE). Li et al. [24] work showed that TFT and TiDE are effective in load forecasting for electricity when using real-world data.

The development of large time series models and large language models (LLMs) adapted for time series analysis is significant, as it has the potential to create new possibilities. This has been demonstrated by recent frameworks [25,26]. Time-LLM has been shown to outperform conventional deep learning architectures in various forecasting benchmarks [27]. Liao et al. [20] evaluated TimeGPT, a transformer pre-trained on 100 billion data points from domains like finance, weather, and energy. While it requires fine-tuning for load prediction, TimeGPT outperforms benchmarks in low-data settings (e.g. 3–7 training days), especially for short horizons. With more target data, task-specific models often perform better. Its performance also declines under distribution shifts and lacks built-in support for exogenous inputs like weather. Time-LLM, which uses patch reprogramming and structured prompting to align time series with natural language, was introduced by Jin et al. [28]. This adapts LLMs to forecast time series. However, Mohammadi et al. [21] found that Time-LLM was no better than domain-specific models such as TFT and DeepAR at predicting energy use based on external factors. Time-LLM also did not benefit as much as expected from having weather and socio-demographic data.

Effective TL depends on the critical process of source selection, employing a variety of methods. These range from similarity measures such as Dynamic Time Warping (DTW) and Maximum Mean Discrepancy (MMD) [5,6,29], to combined metrics such as Wasserstein distance and Maximal Information Coefficient (MIC) [30]. These methods can significantly improve performance by reducing negative transfers.

Static features, feature importance and explainability in transfer learning

Static features that describe building characteristics or derived load-patterns are important for contextualising energy consumption. Chatopadhyay et al. [31] work demonstrated that using contextual features improved the forecast for time series models such as PatchTST or iTransformer. While models such as TFT explicitly incorporate static covariates, their effective use in TL—especially the transfer of knowledge about their meaning—is a new area.

Mohammadi et al. [21] tested LSTM, DeepAR, TFT, and Time-LLM on short-term electricity forecasts using a London smart meter dataset with static and dynamic exogenous features. While these inputs generally improved accuracy, only TFT fully benefited (18.8% SMAPE on stable data), whereas LSTM and Time-LLM showed limited gains. Their analysis found permutation feature importance (PFI) more reliable than attention or SHAP, underscoring the need for robust FI methods in TL. Ferreira et al. [32], Wang et al. [33], and Giacomazzi et al. [34] used TFTs for electricity and wind forecasting, demonstrating that SF sharpens temporal patterns and enhances accuracy. However, a systematic analysis of the isolated influence of SF on the forecast accuracy of TFTs for load forecasting is still lacking.

Neubauer et al. [11,35] demonstrated that FI in the prediction of heating loads is strongly influenced by building parameters. The multi-step models using SHAP and attention mechanisms highlight how lagged and future features affect forecast accuracy. These FI values effectively capture building-specific information. While studies such as Huang et al. [36] use XAI to explain district heating load forecasts, the idea of using these FI values as a transferable component in a TL framework is new. Recent time series language models like Time-LLM and TimeGPT suggest that generalised knowledge reuse is possible.

1.2. Research gap, objective, contributions and research questions

Despite advances in TL for load forecasting, which have mainly focused on fine-tuning models or domain-independent feature representations, the transfer of interpretable insights from source to target buildings remains underdeveloped. In particular, the use of FI values (e.g. SHAP values) and load-pattern metrics as transferable inputs has not been explored, even though these values encode meaningful, building-specific insights, as demonstrated by Neubauer et al. [11,35]. This may offer advantages for transfer learning.

To close this research gap, this study introduces a novel hybrid TL framework that combines feature- and parameter-based TL for short-term heating load forecasting, systematically evaluating the suitability of SHAP values and load-pattern metrics as SF across different models and TL strategies. The analysis focuses on a 24-hour forecast horizon, which is particularly relevant for load flexibility in buildings with limited thermal storage capacities. The primary *objectives* are threefold: (a) to establish and rigorously evaluate an FI-enhanced TL strategy that injects SHAP-derived FI values and load-pattern metrics as SF; (b) to identify the most suitable models for this task under varying data constraints; and (c) to deepen interpretability via a systematic attribution analysis of the TFT.

This FI-enhanced strategy is benchmarked on 18 real-world hourly datasets covering multiple building types and climate conditions, using seven time series models: MLP, LSTM, TFT, PatchTST, TiDE, TimeMixer, and Time-LLM. Additionally, the effectiveness of various transfer-learning strategies—including fine-tuning (FT), fine-tuning with frozen layers (FTF), and a zero-shot (ZS) baseline—is analysed across different levels of target-data availability, an aspect that has thus far been scarcely explored in heating-load forecasting. The *main contributions* are:

1. **FI-enhanced TL methodology**—Formulation of a novel TL framework that tests SHAP-based FI values and load-pattern metrics as building-level static covariates and quantifies their modest, model-dependent impact across architectures and TL strategies.
2. **Large-scale empirical study**—The method is benchmarked on 18 mostly real-world hourly datasets spanning residential, office, and university buildings on two continents, using seven time series models (MLP, LSTM, TFT, PatchTST, TiDE, TimeMixer, and Time-LLM). These models are pre-trained on 108 heating load datasets to enable TL across building types and regions.
3. **Data-scarcity and fine-tuning analysis**—Quantification of performance under varying target-data availability (0% (ZS), 10%/100% (NTL/FT/FTF) and comparison of fine-tuning schemes, including selective layer freezing.

4. **Practical feature engineering for TL**—All transferable features are extracted *solely* from historical load, time features and weather data, making the workflow applicable when detailed building metadata are unavailable.

Based on the identified *research gaps* and the contributions of the paper, the following *research questions* are addressed:

- RQ 1:** To what extent do transfer learning strategies and SHAP-based static features affect short-term forecasting accuracy across diverse buildings, including data-scarce cases?
- RQ 2:** Which transfer learning strategy (fine-tuning, frozen-layer fine-tuning, or zero-shot) and which model performs best across varying data availability levels, and dataset sources?
- RQ 3:** How do model and TL strategy performances vary between seen and unseen buildings and between different training data lengths?
- RQ 4:** What do TFT internal attributions and SHAP-based explanations reveal about the roles of static, past, and future features, across TL scenarios?

2. Methods

This section outlines the methodological approach of the study. First, there is an overview of the principles of TL. Subsequently, the derivation of novel SF, central to this work, is detailed, including FI values from SHAP analysis and quantitative load-pattern characteristics. Finally, the forecast models and the experimental setup for applying and evaluating TL strategies across various data availability scenarios are described, as are the evaluation metrics used. A conceptual overview of the study is presented in Fig. 1.

2.1. Transfer learning principles

TL aims to improve learning in a target domain $\mathcal{D}_T$ with a target task $\mathcal{T}_T$ by leveraging knowledge from a source domain $\mathcal{D}_S$ and a source task $\mathcal{T}_S$, especially when the target data are limited [37]. A domain $\mathcal{D} = \{\mathcal{X}, P(X)\}$ consists of a feature space $\mathcal{X}$ and its probability distribution $P(X)$. A task $\mathcal{T} = \{\mathcal{Y}, f(\cdot)\}$ involves a label space $\mathcal{Y}$ and a predictive function $f(\cdot)$. For heating load forecasting, $\mathcal{X}$ includes temporal and exogenous features, and $\mathcal{Y} \subset \mathbb{R}$ is the heating load. This study focuses on inductive TL, in which labelled target data is used to adapt knowledge—both model parameters and feature-derived insights—learned from source domains.

2.2. Used dynamic and static features

All models predict the hourly heating load ($\dot{Q}_{\text{heat}}$) over a 24-hour forecast horizon (FHor), using lagged hours (LH) of past observations that varies by model. The models incorporate autoregressive features (past values of the heating load itself) alongside dynamic exogenous features, which are available as past $x_{t-\text{LH}+1:t}$ and future(-known) features $x_{t+1:t+\text{FHor}}$. These dynamic exogenous features include outdoor temperature T_{out} , global horizontal irradiance (GHI), hour of the day (Hour), day of the week (Day). The core of the approach is the introduction of two types of static exogenous features x_s transferred to the target models: SHAP-based FI values and load-pattern metrics. All features were scaled using robust scaler to mitigate the influence of outliers and to stabilise training.

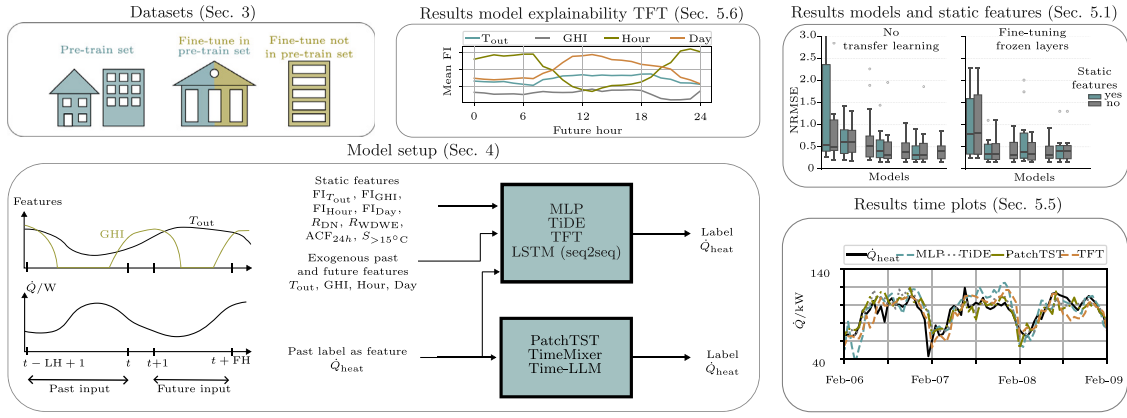


Fig. 1. Overview of the study and used methods.

2.2.1. Feature importance

SHAP [38] values offer a method for determining FI in ML models. These values indicate the extent to which a feature influences the target variable, thereby increasing the explainability of prediction models. The effectiveness of SHAP values in identifying relevant features and evaluating their impact on predictions in energy and load forecasting has been demonstrated in several studies [35,39]. This motivates their use here as a powerful, widely adopted XAI technique. In this work, SHAP is used as a widely adopted feature importance method to encode selected features as static descriptors.

This study uses TreeSHAP to determine SHAP values based on an XGBoost model. For each building, the absolute SHAP contributions of the four dynamic exogenous variables T_{out} , GHI, Day, Week are calculated. These are then normalised so that their sum equals 1, resulting in relative SHAP values. The normalised SHAP values, denoted by $FI_{T_{out}}$, FI_{GHI} , FI_{Hour} , and FI_{Day} , thus represent the generalised relative importance on the forecast of the hourly heating load. While TreeSHAP derives FI values from an external surrogate model (XGBoost) to provide a generalised building-level profile, these may differ from the internal mechanisms of the forecasting models, such as the TFT's attention weights. This auxiliary approach may capture coarse, long-term correlations rather than the task-specific predictive relevance exploited by the downstream deep learning models. However, combining external and internal attribution perspectives enables a more comprehensive assessment of influencing factors.

2.2.2. Load-pattern characteristics

In order to describe typical thermal load-patterns, various key values are calculated based on the heating load and the outside temperature. The aim is to use a purely data-driven approach to record the influence of building usage and possible control strategies, such as night-time temperature reductions or reduced set-point temperatures at weekends, on the load profiles. Exact information about usage and control is often unavailable, so this statistical approach to classification is used instead.

The day-night load ratio R_{DN} records possible night-time reductions and can also provide information about general usage and occupancy times. For example, it can show whether buildings have relatively low loads at night due to a night setback control. Similarly, the weekday-weekend load ratio R_{WDWE} provides information on usage and heating strategies by showing whether the load is reduced at weekends, which is typical of office and administrative buildings. The load ratios take temperature-adjusted loads into account. Autocorrelation at a 24-hour lag ACF_{24h} is used to analyse recurring daily patterns and shows how similar the load curves are on consecutive days. Finally, the share of load above 15°C $S_{>15^\circ\text{C}}$ metric indicates whether a significant proportion of energy consumption occurs at outdoor temperatures above 15°C , suggesting that the energy is also being used for domestic hot water, ventilation loads, or other processes. The mathematical definitions are provided in Appendix B.

Table 2

Overview of selected forecasting models. Exogenous support: P = Past, F = Future-known, S = Static.

Model	Architecture	Forecast type	Exogenous support
MLP [43]	MLP	Direct	P/F/S
TiDE [44]	MLP	Direct	P/F/S
TimeMixer [45]	MLP	Direct	–
LSTM [46]	RNN	Both	P/F/S
TFT [23]	Transformer	Direct	P/F/S
PatchTST [47]	Transformer	Direct	–
Time-LLM [28]	LLM	Direct	–

2.3. Model selection

Seven time series forecasting models were selected for their varying architectures to systematically analyse the impact of model complexity on forecast accuracy and resource efficiency. The models selected had been frequently used in previous work, had achieved good results, or were quite novel (Section 1.1). Four different model architectures were investigated: MLP, RNN, Transformer, and LLM. The models were implemented using the NeuralForecast library [40]. Table 2 provides an overview of the selected models. Classical time series baselines (e.g. ARIMA or linear regression) have been widely studied for building energy forecasting and often underperform modern deep learning models in complex, multi-feature settings [23,41,42]. Since the primary aim of this work is to compare TL strategies across recent neural architectures rather than to re-establish the well-known performance gap between statistical and neural approaches, the benchmark is restricted to neural forecasting models.

Explicit physical constraints are deliberately avoided, and emphasis is placed on unrestricted data-driven design. This decision is based on the assumption that the large, heterogeneous pre-training corpus can already capture a large part of the domain structure. The evaluation of physically informed variants (e.g. PINNs) is reserved for future work as they typically require building-specific physical metadata and state measurements (e.g. room temperatures), which are not accessible for the most datasets.

The forecasting type refers to how models generate predictions over the forecast horizon. There are two types: direct models, which produce forecasts for all time steps in a single forward pass; and recursive models, which generate forecasts one step at a time using the previous forecast as input for the next step. All models were configured to follow a direct forecasting type, producing the entire 24-hour forecast horizon in a single forward pass. This approach was chosen to reduce the error accumulation that can occur with recursive methods. The models differ in their support for exogenous features—such as weather data, calendar effects, or the static features proposed in this work.

Table 3
Overview of the used datasets.

Dataset source	Usage type	Data type
Denmark DHS [49]	Residential DHS	Measurements
Serbia DHS [50]	Residential DHS	Measurements
Germany GF	Residential	Measurements
Germany Hotel	Hotels	Measurements
Germany VDI [11]	Office	Simulated
USA BDG2 [51]	Public	Measurements

These inputs are categorised as past features (available over the lagged hours), future features (known over the forecast horizon), or static features (constant per time series). The integration of static features is architecture-specific: for TFT, static covariates are encoded via the dedicated static pathway (Gated Residual Networks and static variable selection/gating) and provide global context to the temporal layers. For models with SF support in our setup (MLP, TiDE, LSTM), the static covariate vector is provided as a building-level conditioning input. Models without SF support are trained and evaluated without static covariates. This design allows the models that support static inputs to condition forecasts on building-specific characteristics captured by the FI and load-pattern metrics.

3. Datasets

A total of 116 datasets from six sources were considered, of which five were based on real measurements and one on simulations. The datasets cover various building types, usages, climatic conditions, and energy efficiency classes. This enables the model to learn a wide range of load profiles and be evaluated under varying conditions. Table 3 provides a summary of the data sources used, the usage type of the building, and whether the data were measured or simulated. Some of the datasets are values from district heating system (DHS) substations. The resolution of all time series data is hourly. For the datasets, weather data for the location was supplemented via OpenMeteo [48]. The weather data used for the simulation is based on the test reference year data provided by the German Weather Service for Berlin and Stuttgart. All datasets are publicly available, with the exception of the datasets from Green Fusion (GF) and the hotel. The hotel data can be provided upon request. Further information on the datasets can be found in Appendix A. The datasets used, along with their exact names, are listed in Table A.14.

Fig. A.13 shows the distribution of individual features and the label across various dataset sources. A comparison of the average daily load-patterns for February and October is presented in Fig. A.14.

4. Model setup

The available data was divided into three groups: datasets that were used for pre-training, datasets that were used for both pre-training and evaluation (seen), and datasets that were used only for evaluation (unseen). A total of 108 datasets of various types were used for pre-training, comprising a total of 1 874 769 hourly heating load values. The forecast accuracy was then evaluated using 18 datasets, taking into account both seen and unseen cases to test generalisability. A detailed breakdown of how the datasets were assigned to the training and test groups is provided in Table A.14.

All datasets use T_{out} , GHI, Hour and Day as features. The influence of these features can vary depending on the model and dataset source (see Fig. A.13). However, the selected features have been shown in numerous studies to be effective for heating load forecasting [35]. Furthermore, in contrast to features such as wind speed, they are not heavily dependent on local conditions, making them difficult to represent using online weather data. The features are considered both past and future features. In the case of past features, past labels are

also considered. Optionally, static features (e.g. SHAP-based FI) can be used as additional features. Forecasts are generated for hours 1, 6, 12, 18, and 24 (hourly forecasts are omitted to reduce computational and memory demands). Unless stated otherwise (e.g. Section 5.5), evaluations are based on the mean across these forecast hours.

The models are trained and forecasts are generated using a rolling window approach. At each time step t , both the lag and forecast windows are shifted forward by one step. In this study, forecast values are generated directly for the entire forecast horizon and are not calculated recursively or iteratively. A base model for each ML model is first pre-trained on a large, diverse aggregation of source datasets (108 datasets). The experimental design compares four TL strategies:

1. *No transfer learning (NTL)*: models that are trained from scratch without TL.
2. *Fine-tuning (FT)*: fine-tuning all layers of the pre-trained model on the target data.
3. *Fine-tuning freezing (FTF)*: selectively freezing pre-trained layers, with only the selected layers being adjusted on the target data.
4. *Zero-shot (ZS)*: directly applying the pre-trained model to the target data without any additional training.

All approaches are tested with and without the proposed SF, which are used as additional exogenous inputs. The datasets are divided into three sets: 70 % for training, 15 % for validation and 15 % for testing. These splits follow a strict chronological order (training, validation, test) with no shuffling to prevent look-ahead bias. The intermediate validation block provides temporal separation between training and test, supporting a realistic generalisation assessment (Table A.13).

The FI is calculated using the test set. The validation set was used for early stopping to prevent overfitting and for hyperparameter tuning during NTL and FT/FTF training. For pre-training, all epochs were completed, and the model checkpoint with the lowest validation error was selected for transfer. For subsequent fine-tuning (FT/FTF), the learning rate was reduced by a factor of 0.1 to ensure the pre-trained model adapted more stably and precisely to the target data.

The performance of each target building and each variant is evaluated using different proportions of the available training data: 10 % and 100 % for NTL, FT, and FTF and 0 % for the ZS strategy. The reduction in training length means that there are cases where the training set only includes a few heating days (Table A.13, VDI 6007). In the FTF approach, earlier layers that are responsible for general feature extraction (e.g. embedding layers, initial convolution or recursion blocks, and early transformer encoder layers) are usually frozen, while subsequent layers that are more specific to the task are fine-tuned. For example, with Transformer-based models like PatchTST, initial patching and embedding layers were frozen to learn general patterns, while subsequent encoder blocks were fine-tuned to adapt to the target domain. Table C.16 shows the frozen layers for the different model architectures.

4.1. Evaluation metrics

The forecast accuracy is assessed using the normalised mean absolute error (NMAE) and the normalised root mean squared error (NRMSE). Let the evaluated forecast hours (FH) be $\{1, 6, 12, 18, 24\}$. Here, $y_{i,h}$ denotes the observed heating load for forecast origin i at forecast hour $h \in \{1, 6, 12, 18, 24\}$, while $\hat{y}_{i,h}$ is the corresponding forecast. Both metrics are normalised by the mean observed load $\bar{y}$ (computed over all $i = 1, \dots, N$ and $h \in \{1, 6, 12, 18, 24\}$ of the evaluation set), enabling direct comparison across datasets with different load levels. To summarise the overall performance across the evaluated forecast hours, the mean NMAE and NRMSE are used.

$$\text{NMAE} = \frac{1}{5N} \sum_{h \in \{1, 6, 12, 18, 24\}} \sum_{i=1}^N \frac{|y_{i,h} - \hat{y}_{i,h}|}{\bar{y}} \quad (1)$$

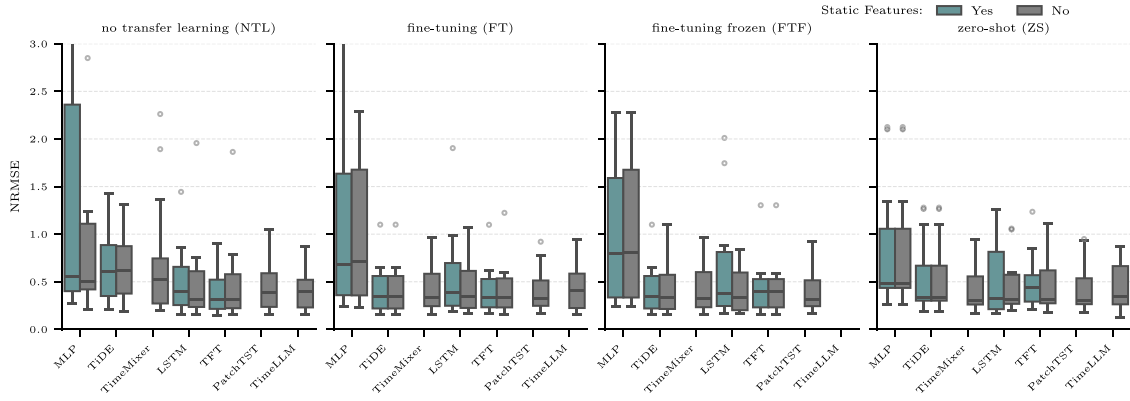


Fig. 2. Distribution of NRMSE depending on the forecast model and various TL approaches (NTL, FT, FTF and ZS), with and without SF. The y-axis is truncated at 3 for visual clarity.

$$\text{NRMSE} = \frac{1}{y} \sqrt{\frac{1}{5N} \sum_{h \in \{1,6,12,18,24\}} \sum_{i=1}^N (y_{i,h} - \hat{y}_{i,h})^2} \quad (2)$$

Performance differences are additionally assessed for statistical significance across datasets. Comparisons between runs with and without SF on the same datasets, as well as paired comparisons between TL strategies (e.g. FT vs. NTL, FTF vs. NTL, ZS vs. NTL), are evaluated using the Wilcoxon signed-rank test. A p -value below 0.05 indicates statistical significance.

4.2. Hyperparameter tuning

A hyperparameter tuning was conducted to determine the optimal settings for the models in use. Various hyperparameters were considered, including input/past hours, batch sizes and the maximum number of training steps. The exact search ranges and optimal values found in each case are summarised in Table C.15. Figs. C.15 and C.16 illustrate the impact of various hyperparameters on the validation loss. Due to the significant resource requirements, hyperparameter tuning was not performed for the Time-LLM. The intention was also to find out if the baseline Time-LLM is better than, or equivalent to, the other models [27].

The tuning was performed using Optuna [52], a Bayesian hyperparameter optimisation framework, with 100 independent runs for each model. Throughout, the mean square error was used as the loss function. Early stopping was employed for NTL, FT, and FTF, with the selected maximum training steps and search ranges varying between the models Appendix C. The hyperparameter tuning was carried out using dataset 490 from the Denmark DHS data source. This dataset was used as a development set to keep the benchmark computationally feasible and to establish a fixed, universal hyperparameter configuration. These configurations are then applied without further re-optimisation across 18 heterogeneous evaluation datasets, providing a consistent assessment of their cross-dataset behaviour. A separate search was performed for each of the four main variants (NTL, FT, NTL+SF and FT+SF), and the best parameters were adopted based on the validation data.

5. Results

The analysis of the results is divided into the following sections. Section 5.1 analyses the effect of TL and SF on forecast accuracy, while Section 5.2 evaluates it across different data sources. Section 5.3 examines the influence of the training fraction on model accuracy, and Section 5.4 examines the effect of whether the test datasets were included in the pre-training set (seen/unseen). Finally, Section 5.5 presents case studies with time series forecasts for selected datasets. Section 5.6 analyses model explainability using the TFT FI.

5.1. Impact of transfer learning strategy and static features on model accuracy

Fig. 2 shows the distribution of NRMSE for the different TL variants—NTL, FT, FTF, and ZS inference—differentiated according to the models used, with and without SF considerations. The use of past, future, and static features for each model followed the exogenous support listed in Table 2. The median values and percentage improvements for all models across the various TL variants are summarised in Table 4.

Overall, the results show that TL significantly improves forecast accuracy. The median NRMSE across all models decreases from 0.445 for NTL to 0.367 for FT, corresponding to an improvement of 17.5%. With FTF, this value decreases further to 0.347, representing an improvement of 22.0% compared to NTL. ZS achieves a median NRMSE of 0.356, which corresponds to a 20.0% improvement over NTL.

At the model level, PatchTST achieves the best result under FT with a median NRMSE of 0.322—an improvement of around 16.4% over NTL. TiDE and TimeMixer also benefit significantly from FT, reducing their NRMSE by 43.7% and 36.0%, respectively. Conversely, LSTM deteriorates under FT from 0.315 to 0.349, corresponding to a −10.8% decrease in forecast accuracy, indicating that unrestricted adaptation of all layers is not effective here. TFT also worsens slightly under FT, with its error increasing from 0.316 to 0.338 (−7%).

The advantages of the FTF strategy in the context of TL are even more pronounced. For FTF, PatchTST achieves the best forecast accuracy with a median NRMSE of 0.315—an improvement of around 18.2% over NTL. TiDE and TimeMixer also improve significantly over their NTL baseline values, with gains of about 46.6% and 38.3%, respectively. MLP continues to have the highest error, which worsens under FTF to 0.803. For TFT, FTF performs significantly worse than FT, with no improvement.

ZS performs well in many models with good forecast accuracy, even without fine-tuning the model on the target dataset. TimeMixer achieves the best overall results with a median NRMSE of 0.300—an improvement of around 42.2% over NTL. PatchTST and TiDE also benefit from ZS, improving by around 21.0% and 45.3%, respectively. Remarkably, even without training on the target dataset, models such as Time-LLM and TFT remain competitive, with NRMSE values of 0.346 and 0.316, respectively.

The influence of SF, i.e., SHAP-based FI and load-pattern metrics, varies considerably depending on the model and the TL strategy. For models with and without SF (LSTM, MLP, TiDE and TFT), forecast accuracy improves by 11.2% (median NRMSE from 0.485 to 0.430) under NTL. Under FT, the improvement is marginal at 0.1%, while under the FTF the NRMSE increases from 0.369 to 0.383 (a forecast accuracy degradation of 3.7%). At the individual model level, TiDE remains largely unaffected by SF in all variants. TFT shows a slight

Table 4

Median NRMSE values and the relative improvements in % compared to the baseline (NTL or without SF). Positive values indicate a reduction in error (i.e., improvement); negative values indicate performance degradation. Best value per NRMSE column is highlighted in bold. The bottom row, only SF m., aggregates medians over the subset of models that support static features.

Model	NTL	FT	FTF	ZS	NTL +SF	FT +SF	FTF +SF	ZS +SF	FT vs NTL	FTF vs NTL	ZS vs NTL	NTL+SF vs NTL	FT+SF vs FT	FTF+SF vs FTF	ZS+SF vs ZS
MLP	0.505	0.718	0.803	0.484	0.553	0.684	0.795	0.484	-42.2	-59.0	+4.2	-9.5	+4.7	+1.0	0.0
TiDE	0.616	0.347	0.329	0.337	0.608	0.347	0.347	0.337	+43.7	+46.6	+45.3	+1.3	0.0	-5.5	0.0
TimeMixer	0.519	0.332	0.320	0.300	-	-	-	-	+36.0	+38.3	+42.2	-	-	-	-
LSTM	0.315	0.349	0.340	0.318	0.395	0.388	0.373	0.328	-10.8	-7.9	-1.0	-25.4	-11.2	-9.7	-3.1
TFT	0.316	0.338	0.402	0.316	0.309	0.330	0.402	0.435	-7.0	-27.2	0.0	+2.2	+2.4	0.0	-37.7
PatchTST	0.385	0.322	0.315	0.304	-	-	-	-	+16.4	+18.2	+21.0	-	-	-	-
Time-LLM	0.398	0.409	-	0.346	-	-	-	-	-2.8	-	+13.1	-	-	-	-
Median	0.445	0.367	0.347	0.356	0.430	0.373	0.383	0.435	+17.5	+22.0	+20.0	+3.4	-1.6	-10.4	-22.2
only SF m.	0.485	0.373	0.369	0.393	0.430	0.373	0.383	0.435	+23.0	+23.9	+18.9	+11.2	+0.1	-3.7	-10.8

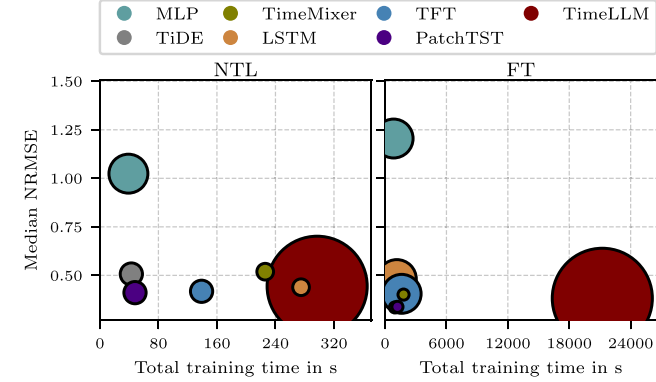


Fig. 3. Relationship between total training time, median NRMSE, and memory requirements for the NTL and FT strategy.

improvement under FT+SF (from 0.338 to 0.330), but no change under FTF+SF.

Overall, the statistical tests (Section 4.1) indicate no consistent, statistically significant benefit from SF on forecast accuracy across datasets for any TL variant (NTL, FT, FTF, ZS). Specifically, paired comparisons between runs with and without SF show no significant differences for most model and TL combinations. The only significant case is a degradation for LSTM under FT (Table D.17). In contrast, several paired TL strategy comparisons versus NTL are significant across datasets despite sometimes small median changes, because the tests quantify the consistency of improvements across datasets. For example, TiDE shows significant improvements under both FT and FTF ($p = 0.0003$), whereas other architectures show weaker or non-significant effects.

Trade-off between model accuracy, computation time, and memory usage

Table 5 (NTL) and Table 6 (FT) provide an overview of the number of parameters, memory requirements, and runtime.¹ Fig. 3 shows this relationship graphically. It is evident that achieving good forecast accuracy does not necessarily require significantly more resources or training time. The number of parameters and storage sizes differ between the NTL and FT setups because hyperparameters were tuned separately for the two datasets. The variations in input length, hidden layers, and architecture-specific configurations—resulting from separate hyperparameter tunings for NTL and FT—lead to differences in the number of parameters and storage sizes.

PatchTST achieves the lowest median NRMSE among all models under FTF (0.322, 18.2% over NTL), with a fine-tuning time of only 663 s and very low storage requirements of 0.12 MB. By comparison, Time-LLM attains a median NRMSE of 0.409 (about 27% higher than

Table 5

Overview of the number of parameters, storage size, training duration, and forecast duration for the different models with NTL.

Model	Number of parameters	Storage in MB	Train in s	Forecast in s
MLP	1.62×10^6	3.08	39	797
TiDE	2.41×10^5	0.46	43	824
TimeMixer	1.20×10^5	0.23	226	866
LSTM	8.80×10^4	0.17	275	871
PatchTST	1.60×10^5	0.30	49	828
TFT	1.24×10^5	0.28	166	810
Time-LLM	5.39×10^7	340.12	297	1651

Table 6

Overview of the number of parameters, storage size, pre-train duration, fine-tune duration, and forecast duration for the different models with FT.

Model	Number of parameters	Storage in MB	Pre-train in s	FT in s	Forecast in s
MLP	1.39×10^6	2.66	241	596	977
TiDE	8.06×10^4	0.15	299	666	1019
TimeMixer	5.93×10^4	0.12	1037	770	1063
LSTM	1.39×10^6	2.66	423	729	1066
PatchTST	6.51×10^4	0.12	342	663	982
TFT	1.64×10^6	3.13	1262	954	1161
Time-LLM	5.39×10^7	340.12	19 899	1332	2691

PatchTST) while requiring 2833× more storage and 2.01× more fine-tuning time. TimeMixer also performs well under FTF, reaching a median NRMSE of 0.320 (38.3% over NTL) but requiring 770 s for fine-tuning. TiDE follows closely with an NRMSE of 0.329, fine-tuning in 666 s, and a storage size of only 0.15 MB, making it very resource-efficient. Despite 1.4×10^6 parameters, MLP is clearly outperformed by specialised architectures, and Time-LLM demands the most resources (340 MB, 1332 s) while delivering a median NRMSE of 0.409, about 27% higher than PatchTST.

It is notable that fine-tuning is more time-consuming than pure pre-training for some models, primarily because the learning rate is deliberately reduced by a factor of 0.1 during fine-tuning. This deliberate reduction slows convergence and typically increases training time, but enables more stable adaptation to the target data and often results in improved forecast accuracy.

Overall, the results confirm the benefits of TL, particularly when using FTF, in improving the accuracy of the forecast. TiDE achieves a good trade-off between runtime, memory usage, and accuracy, demonstrating robust performance across all transfer learning strategies. PatchTST is particularly well-suited to applications requiring very short training times and stable forecast accuracy. Additionally, ZS is a powerful alternative as it requires no training in the target region and still offers good accuracy. This makes ZS an attractive option for applications where resources or data are limited. For applications with limited resources, lightweight architectures such as PatchTST or TiDE, combined with FTF or ZS, offer a favourable balance between accuracy, memory usage, and training time.

¹ All simulations were performed on an NVIDIA RTX 4090 GPU, using float 16 weight precision.

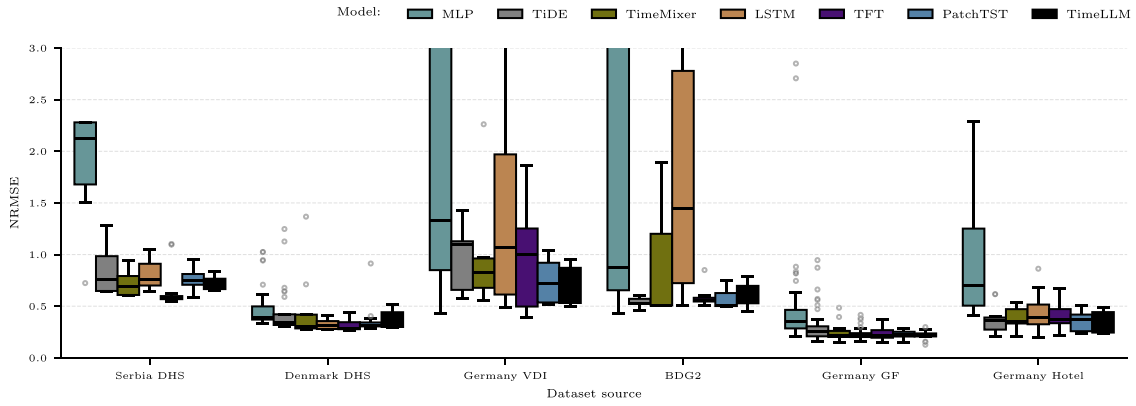


Fig. 4. Distribution of NRMSE depending on the ML models and dataset source used. The median values are shown above the individual box plots. The y-axis is truncated at 3 for visual clarity. The results include the TL strategies NTL, FT, FTF, and ZS.

Table 7

Overall ranking of models based on their median NRMSE across all datasets.

Model	Sum of rankings	Mean ranking
PatchTST	16	2.67
TiDE	19	3.17
Time-LLM	19	3.17
TFT	20	3.33
TimeMixer	21	3.50
LSTM	34	5.67
MLP	39	6.50

5.2. Impact of different dataset source and transfer learning strategy on model accuracy

Fig. 4 shows the distribution of NRMSE across different models and datasets. The median NRMSE values vary substantially depending on the dataset. The Green Fusion dataset achieves the best forecasts, with a median NRMSE of approximately 0.23, while the VDI 6007 reference building shows the highest errors, averaging around 0.93. This highlights that forecast accuracy depends not only on the model but also heavily on the characteristics of the underlying time series. Some load profiles are easier to learn than others (Section 5.5), and the features can provide a more accurate representation of the heating load.

A comparison of the best-performing models in each dataset shows that PatchTST and TiDE most frequently achieve the lowest median NRMSE, each ranking first in two datasets. Time-LLM and TimeMixer lead in one dataset each and also show strong overall performance. PatchTST achieves the best total ranking score (16), followed closely by TiDE and Time-LLM (19 each). In contrast, LSTM ranks lower, and MLP performs worse, especially on the Serbia DHS dataset, where it reaches a median NRMSE of over 2.1.

Table 7 summarises the overall model rankings (1–6 per dataset) across all datasets. PatchTST achieves the best average rank (2.67), followed by TiDE and Time-LLM (3.17 each). TFT and TimeMixer follow with slightly higher average ranks, while LSTM and MLP clearly lag behind.

In summary, the choice of the best model depends on the specific dataset. PatchTST and TiDE deliver strong, consistent results across a variety of time series and achieve the best overall rankings. TimeMixer and Time-LLM also perform competitively, particularly when the trade-off between accuracy, training time, and memory usage is considered.

5.3. Impact of training set length and transfer learning strategy on model accuracy

Fig. 5 shows the impact of varying training fractions on median NRMSE values for the variants NTL, FT, and FTF. The training fraction is the proportion of the maximum available training data used during model training. The shorter training periods also mean that some training periods (see VDI 6007) are largely outside the heating period (Table A.13). Table 8 lists the corresponding median values. Reducing the training fraction from 1.0 to 0.1 increases the median NRMSE by 37.2% under NTL and by 66.5% under FT. For FTF, the increase is even higher at 79.7%. Despite its strong median performance at larger training fractions, FTF degrades sharply under very limited target data for several architectures (e.g. MLP, LSTM), suggesting that extensive freezing can restrict adaptation when only few target samples are available. In this low-data scenario, ZS can therefore be the more robust option.

The models react very differently to reduced training data. MLP and LSTM deteriorate significantly under NTL and FT (up to +206.3% and +112.7%, respectively), indicating a strong dependence on the amount of data. In contrast, PatchTST remains remarkably stable across all variants: under NTL, the error slightly decreases, while under FT and FTF, the values increase only marginally. TiDE exhibits similar behaviour, showing a clear improvement under NTL and stable values under FT and FTF. TFT remains largely constant under FTF, while Time-LLM benefits from having less data under FT, improving forecasts by 20.8%.

Overall, ZS performs well, achieving a median NRMSE of 0.381 without any training. In several cases, fine-tuning with just 10% of the training data achieves comparable or better performance. TiDE and Time-LLM demonstrate clear advantages with FT in this low-data scenario, with TiDE showing further improvement with FT and FTF. This suggests that these architectures can effectively adapt to task-specific patterns even with limited data. In contrast, PatchTST shows no improvement over ZS with 10% training data, suggesting that its pre-trained weights already capture most of the relevant temporal dependencies. ZS achieves the same or better performance than FT/FTF for TimeMixer, LSTM, and TFT, which points to the robustness of pre-trained representations and the potential risk of overfitting when adapting to small datasets. Some models perform better with 10% of the training data than with 100%, which can be attributed to seasonality of the data or possible overfitting of the models. These results show that the benefit of fine-tuning depends heavily on the model architecture. ZS often performs well and is therefore a competitive choice.

Table 8

Effect of reducing the training fraction from 1.0 to 0.1 on the median NRMSE across different models and transfer learning strategies. Positive values indicate a reduction in error (i.e., improvement); negative values indicate performance degradation.

Model	NTL			FT			FTF			ZS
	1.0	0.1	0.1 vs 1.0 (%)	1.0	0.1	0.1 vs 1.0 (%)	1.0	0.1	0.1 vs 1.0 (%)	0.0
MLP	0.507	1.553	−206.3	0.687	1.452	−111.4	0.803	1.683	−109.5	0.484
TiDE	0.616	0.392	+36.4	0.347	0.325	+6.3	0.329	0.310	+5.8	0.337
TimeMixer	0.519	0.525	−1.2	0.332	0.514	−54.8	0.320	0.520	−62.5	0.300
LSTM	0.370	0.629	−70.0	0.369	0.785	−112.7	0.358	0.847	−136.4	0.328
TFT	0.309	0.395	−27.8	0.335	0.406	−21.2	0.402	0.406	−1.0	0.359
PatchTST	0.385	0.368	+4.4	0.322	0.340	−5.6	0.315	0.336	−6.7	0.304
Time-LLM	0.398	0.468	−17.6	0.409	0.324	+20.8	–	–	–	0.346
Median	0.439	0.602	−37.2	0.369	0.613	−66.5	0.365	0.657	−79.7	0.381

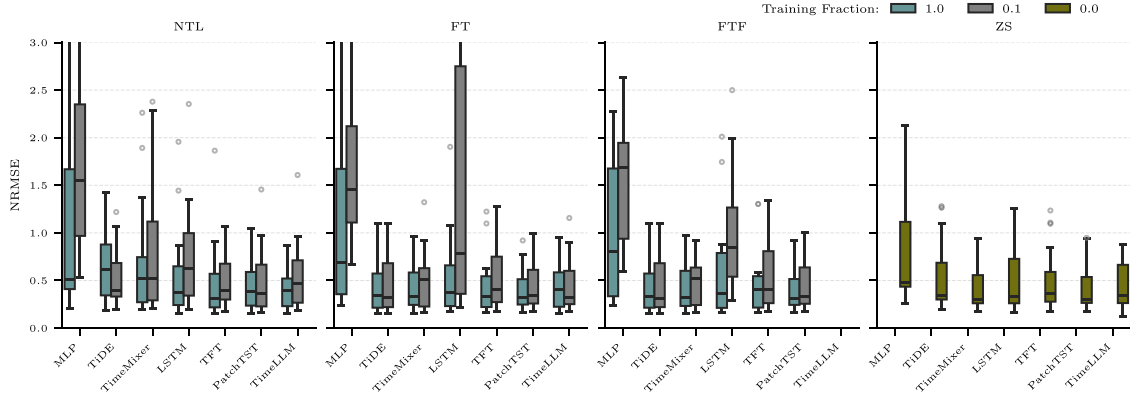


Fig. 5. Distribution of NRMSE depending on ML models, transfer learning strategy, and training fraction. The y-axis is truncated at 3 for visual clarity.

5.4. Impact of test set seen/unseen and transfer learning strategy on model accuracy

The generalisation capability is assessed by distinguishing between test datasets that were included during pre-training *seen* and those that originate from a previously unknown domain *unseen*. Since absolute NRMSE values for the seen/unseen variants depend on dataset-specific properties such as variance and complexity, the evaluation focuses on relative improvements over the baseline variant NTL. Fig. 6 shows the NRMSE distribution for all TL strategies, grouped by pre-trained dataset (seen and unseen) and SF. The underlying results apply to all models and are limited to the training fractions (0 for NTL and 1 for FT and FTF). The median NRMSE is reported for each variant. The lowest NRMSE are coloured red.

Table 9 summarises the relative improvements over the NTL baseline for FT, FTF, and ZS, along with the performance difference Δ_{impr} between seen and unseen datasets. Here, Δ_{impr} reflects the difference in improvement rates, not absolute errors, and indicates how consistently a TL strategy performs across domains. Smaller absolute Δ_{impr} values imply better generalisation.

The results show that all TL strategies tend to improve forecast accuracy compared to the NTL baseline. For models without SF, FTF achieves a strong improvement of 14.6% on seen datasets and, interestingly, performs even better on unseen datasets (21.9%, $\Delta = -7.3\%$). By contrast, the ZS strategy is more balanced, with improvements of 11.1% (seen) and 4.6% (unseen), confirming its robustness ($\Delta = 6.5\%$). Small absolute Δ values indicate that a strategy generalises well, i.e., performs similarly on both previously seen and unseen domains; larger Δ values signal increased sensitivity to domain differences.

At the model level, generalisation gaps Δ_{err} (Table D.19)—defined as the difference in median NRMSE-100 between unseen and seen domains—vary considerably across architectures. Here, smaller Δ_{err} values indicate higher robustness. The lowest gaps occur for TFT under FT ($\Delta_{\text{err}} = -0.01$) and LSTM under FTF ($\Delta_{\text{err}} = 4.76$) and ZS ($\Delta_{\text{err}} =$

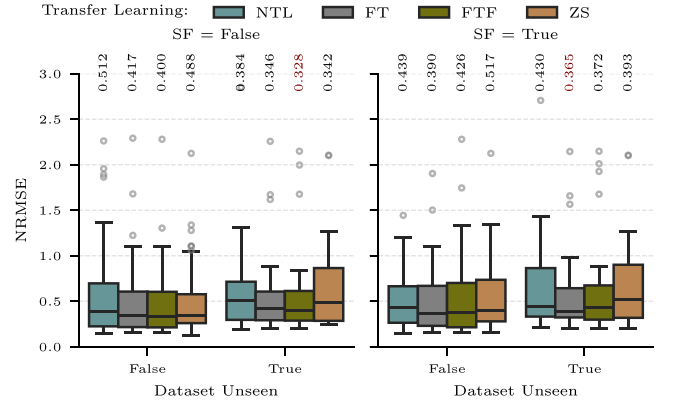


Fig. 6. Distribution of NRMSE by dataset inclusion in pre-training (seen vs. unseen) and use of static features. The y-axis is truncated at 3 for visual clarity. The median values are above the individual boxes.

Table 9

Relative improvements (% vs. NTL) for seen and unseen datasets grouped by TL strategy and SF usage. The difference Δ_{impr} indicates generalisation, where smaller absolute values imply better robustness across domains.

Dataset	SF	FT vs NTL	FTF vs NTL	ZS vs NTL
Seen	No	10.1	14.6	11.1
Unseen	No	18.5	21.9	4.6
Δ (seen–unseen)	No	−8.4	−7.3	+6.5
Seen	Yes	15.2	13.6	8.6
Unseen	Yes	11.1	2.9	−17.7
Δ (seen–unseen)	Yes	+4.1	+10.7	+26.3

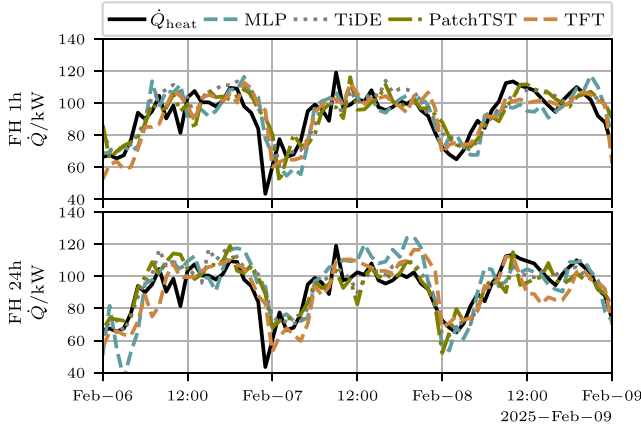


Fig. 7. Time course of heating load and 1 h and 24 h forecasts for the MLP, TiDE, PatchTST, and TFT models with FTF for a GF dataset over three days.

9.72), suggesting that these models generalise most consistently across domains. In contrast, large gaps for MLP under ZS ($\Delta_{\text{err}} = 18.06$) or Time-LLM under ZS ($\Delta_{\text{err}} = 20.77$) indicate strong sensitivity to unseen data. Negative gaps (e.g. MLP under FT) imply slightly better performance on unseen datasets but are likely due to dataset variability rather than genuine robustness.

These differences are likely due to the interaction between model architectures and heating load characteristics (Fig. A.14). Models like TFT and LSTM may better capture transferable temporal dependencies, while TimeMixer's convolutional sequence modelling also yields stable performance. Conversely, feature-dependent models with low sequence inductive bias (e.g. MLP) and parameter-heavy, pretraining-dependent models (e.g. Time-LLM) show larger gaps. Overall, generalisation performance varies greatly depending on the model and transfer strategy used.

5.5. Time series forecast evaluation

The forecast accuracy depends not only on the selected model (Section 5.1) and the quantity and quality of the training data (Section 5.3), but also significantly on the properties of the label itself. These include, for example, the load profile, its pattern, and possible measurement noise or errors.

Fig. 7 shows an example of the course of the heating load and the forecasts for three days for the MLP, TiDE, PatchTST, and TFT models in a GF dataset. The models were trained with the FTF strategy. The upper part of the figure shows the 1-hour forecast, and the lower part shows the 24-hour forecast. The NRMSE for the individual forecast hours are listed in Table 10.

So far, the evaluation metrics have been considered in relation to the first, sixth, twelfth, eighteenth, and twenty-fourth hours of the forecast. Next, the forecast hours will be considered individually. The forecast accuracy decreases as the forecast horizon increases, particularly for MLP and TiDE. PatchTST, on the other hand, shows remarkable stability; the error only increases slightly from 1 h to 24 h (ca. 8 % for FTF and 3.3 % for ZS). In comparison, Fig. 8 shows the same models using the ZS strategy. The forecasts are smoother and show less variation, indicating a reduced fit of the model to the target time series.

Model differences are even more apparent in the Serbia L8 dataset, which has poorer forecast accuracy (Figs. 9 and 10). The load profiles in the night-time and afternoon hours show significant reductions in load to 0 kW. These sharp reductions are not fully captured by most models. Table 11 shows that TiDE greatly increases the error over the forecast horizon for NTL (144 %), but remains significantly more robust for FTF (8.8 %).

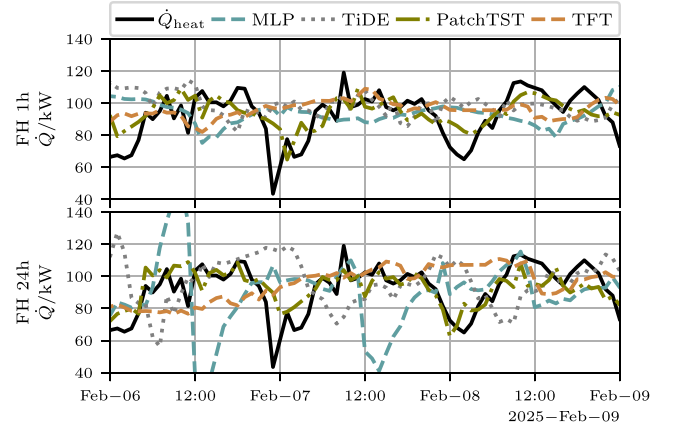


Fig. 8. Time course of heating load and 1 h and 24 h forecasts for the MLP, TiDE, PatchTST, and TFT models with ZS for the GF dataset over three selected days.

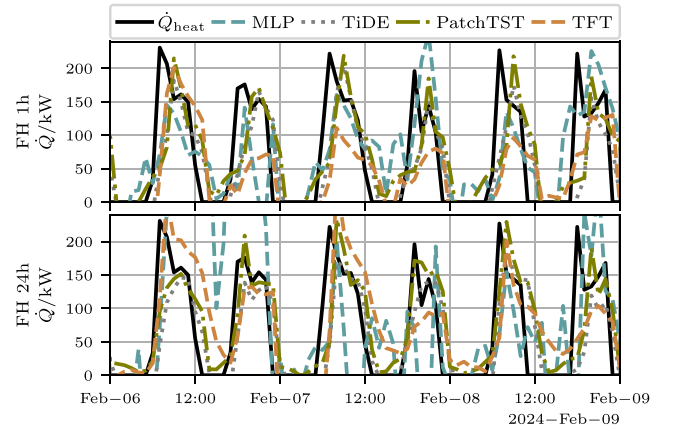


Fig. 9. Time course of heating load and 1 h and 24 h forecasts for the MLP, TiDE, PatchTST, and TFT models with FTF for the Serbia DHS L8 dataset over three days.

Table 10

Median NRMSE at FH 1 h and 24 h for a GF dataset, including relative change.

Model	NTL			FTF			ZS		
	1 h	24 h	Δ (%)	1 h	24 h	Δ (%)	1 h	24 h	Δ (%)
MLP	0.261	0.322	+23.4	0.220	0.259	+17.6	0.397	0.480	+20.9
TiDE	0.446	0.384	−13.9	0.188	0.205	+9.0	0.313	0.361	+15.3
TFT	0.184	0.188	+2.2	0.180	0.200	+11.1	0.266	0.262	−1.5
PatchTST	0.210	0.216	+2.9	0.199	0.215	+8.0	0.211	0.218	+3.3

Table 11

Median NRMSE at FH 1 h and 24 h for the Serbia L8 dataset, including relative change.

Model	NTL			FTF			ZS		
	1 h	24 h	Δ (%)	1 h	24 h	Δ (%)	1 h	24 h	Δ (%)
MLP	0.936	0.935	−0.1	1.305	2.076	+59.1	2.074	2.642	+27.4
TiDE	0.844	2.056	+143.7	0.773	0.841	+8.8	1.535	1.462	−4.8
TFT	0.555	0.650	+17.1	0.518	0.622	+20.1	1.131	1.101	−2.7
PatchTST	0.502	0.578	+15.1	0.592	0.579	−2.2	0.626	0.652	+4.2

PatchTST precisely maps the load profiles over different time horizons under both FTF and ZS. The difference between 1 h and 24 h forecasts remains small (Tables 10 and 11), which indicates the robustness of the model structure. The stability of PatchTST across different forecasting horizons can be explained by its patch-based processing, which integrates local and global pattern data to reduce errors in long-term forecasts. Without domain-specific fine-tuning, ZS models

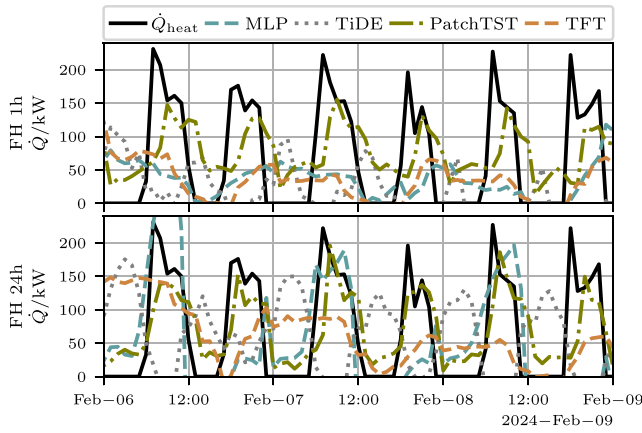


Fig. 10. Time course of heating load and 1 h and 24 h forecasts for the MLP, TiDE, PatchTST and TFT models with ZS for the Serbia DHS L8 dataset over three days.

primarily predict mean values, which limits variability and makes them less responsive to dynamic changes, resulting in smoothing.

Although ZS also often achieves low NRMSE values across all models (Fig. 2), these are partly based on a strong smoothing and centring of the forecasts (see Fig. 8). This is most pronounced for MLP, but also affects generally strong performers like TiDE when faced with abrupt load transitions, as observed for the Serbia L8 dataset (Fig. 10, Table 11). Consequently, this leads to limited responsiveness to dynamic load-patterns. FTF, on the other hand, allows a targeted adaptation to such structures and shows further improvements, especially for models with an already good forecast accuracy (e.g. PatchTST or TiDE).

5.6. Model explainability for the TFT model

The model explainability of the TFT model allows for a detailed analysis of how individual features influence the forecast. Unlike model-independent methods such as PFI, this model-specific attribution method is based on the TFT's internal weightings, for example, through its gating and selection mechanisms.

The features are divided into past, future, and static features. Fig. 11 shows the distribution of the TFT FI across various datasets for the NTL variant. Its value differs greatly depending on the dataset. This indicates that the heating load is strongly influenced by the various features to different extents.

The mean relative TFT FI values across all datasets (Table 12) show relatively similar contributions for past and future features, with values between 0.164 and 0.280. A strong dependency of the heating load on the hour of day does not necessarily result in a high FI for the feature Hour. For example, in the Serbia DHS dataset, the load strongly depends on the hour (Fig. A.14), yet the model assigns only a comparatively low FI to the Hour feature (Fig. 11).

Among the SF, autocorrelation at 24 h ($ACF_{24} = 0.227$) and the day-night quotient ($R_{DN} = 0.202$) show particularly high contributions. The TFT's model-internal FI sometimes differs significantly from TreeSHAP FI (see Fig. A.13). Previous studies (e.g. Neubauer et al. [35]) have shown that DeepSHAP often identifies different influence structures in LSTM-based seq2seq models than those identified in this study.

The relevance of past load values increases systematically, rising from 0.218 (NTL) to 0.773 (FT) and 0.913 (FTF). This indicates that the autoregressive component is becoming more important, while the influence of exogenous features is decreasing. A complete overview of all FI for the different TL strategies is provided in Table D.18.

This equalisation of the FI is evident when the FI values are analysed according to the input hours. In addition to the total FI across all past and future hours, its significance can be examined on a per-input-step

Table 12

Mean relative TFT FI from the explainability analysis of the TFT model, grouped by feature type (future, past, static). SHAP-based SF are denoted by FI.

Feature	Past	Future	Static
T_{out}	0.203	0.255	0.056 ($FI_{T_{out}}$)
GHI	0.220	0.216	0.095 (FI_{GHI})
Hour	0.195	0.280	0.185 (FI_{Hour})
Day	0.164	0.250	0.042 (FI_{Day})
$\dot{Q}_{heat}$ (past load)	0.218	–	–
R_{DN}	–	–	0.202
R_{WDWE}	–	–	0.104
$ACF_{24\ h}$	–	–	0.227
$S_{>15^{\circ}C}$	–	–	0.088

basis for each hour. However, there are some indications regarding future futures input hours. Fig. 12 shows the mean TFT FI over all datasets for future hours, with NTL shown in the upper subplot and FT and FTF in the lower subplot. For the FT variants, the FI remains almost constant over the entire forecast horizon, likely due to averaging across datasets and the uniform patterns learned during pre-training. However, for NTL, it decreases significantly between 6 and 22 h. During this period, the importance of the other features increases, indicating a more differentiated use of future features.

The strong influence of past load is evident not only in the TFT but also in models without exogenous features (Table D.18), such as PatchTST and Time-LLM. In some cases, these models perform better than models with exogenous features, which highlights past values of the label represent a dominant source of information—even for models that exclusively use historical labels as input, such as PatchTST and Time-LLM.

5.7. Summary of key findings

The main findings of this study in relation to the research questions and other key findings can be summarised as follows:

- RQ 1:** Adding SHAP-based FI and load pattern metrics as SF had only a limited impact on forecast accuracy. In some cases, they slightly improved results; in others, they worsened them. Architectures designed for SF (e.g. TFT) showed advantages in some cases.
- RQ 2:** Fine-tuning strategies strongly influenced performance. FTF reduced median NRMSE by 22.0% compared to NTL, with TiDE achieving improvements of up to 46.6% in the best case. With only 10% training data, performance declined for most models, except PatchTST and TiDE, which showed only minor increases of 6.7% or improvements of –5.8%, respectively.
- RQ 3:** TL improves over NTL for both seen and unseen buildings; FTF gives the largest gains, while ZS is most robust—on unseen data it still reduces median NRMSE by 4.6% vs. NTL. With only 10% training data, ZS remains competitive and PatchTST/TiDE degrade least.
- RQ 4:** TL shifts TFT focus strongly towards past loads: TFT FI for past load increases from 0.218 (NTL) to 0.773 (FT) and 0.913 (FTF); internal TFT FI diverges from TreeSHAP.

- PatchTST and TiDE achieved the best trade-off between accuracy and efficiency.
- Fine-tuned sequence-based models like PatchTST and TimeMixer matched or outperformed more complex architectures using static and future inputs (e.g. TFT), demonstrating their strength for TL-based heating load forecasting.

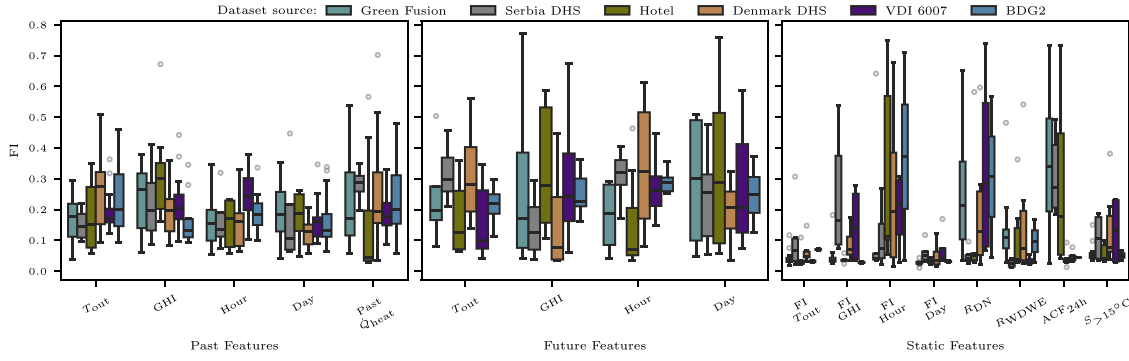


Fig. 11. TFT FI based on the categories past, future, and static features for the various datasets.

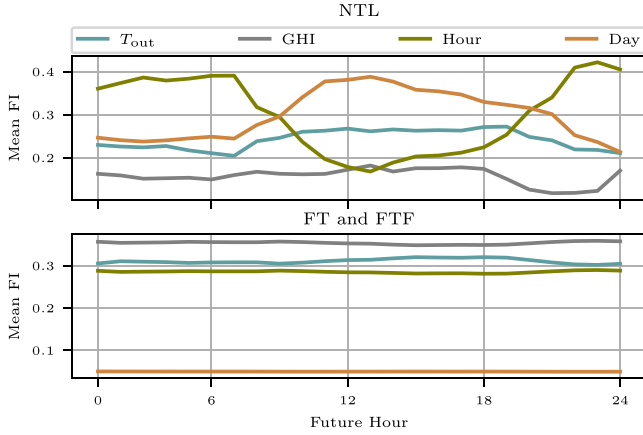


Fig. 12. Mean TFT FI as a function of future hours for the NTL, FT, and FTF variants.

Overall, these results show that targeted TL strategies can produce fast and stable heating load forecasts, even when data is limited. Using additional SF based on FI did not often provide a significant advantage, suggesting that essential patterns are already present in the temporal input series.

6. Discussion

This section discusses the findings and outlines current limitations and directions for future research.

6.1. Focus on median-based evaluation

Most evaluations of models, TL strategies and static features are based on the median value. This methodological decision is due to the comprehensive and holistic nature of this study, which incorporates a large number of datasets, models, TL strategies, reduced training sets and model explainability analyses. Median-based evaluation makes it possible to identify overarching trends and robust patterns that might otherwise be obscured by dataset-specific characteristics in more detailed, individual case analyses. This approach prioritises insights into general trends, which are more relevant to the development of robust TL strategies than analysing individual cases in detail. Furthermore, Section 5.5 presents exemplary time series for selected models and datasets, as well as various forecast horizons, to illustrate the findings using concrete examples.

6.2. Comparison with related work

The results of this study are compared to prior research. In Li et al. [24], TiDE and TFT were also identified as powerful models for load

forecasting. Gao et al. [16] reported a 19.7 percentage point reduction in MAPE by fine-tuning a seq2seq model. In contrast, our results show no improvement from FT for the LSTM-based seq2seq model. However, on average across all models, FT yields large gains, with an average improvement (NRMSE) of 17.5 %.

In a study by Spencer et al. [14], PatchTST achieved a 15.9 % improvement in MAE through FT. Our experiments also showed clear improvements for PatchTST (NRMSE 16.4 % for FT and 18.2 % for FTF). Other models, such as TimeMixer (36.0 %) and TiDE (43.7 %) performed even better under FT.

6.3. Why zero-shot learning delivers such good results

Although ZS methods have received less attention to date, Liao et al. [20] demonstrate that large pre-trained models can be competitive even without target data. Our results confirm this; on average, ZS reduced the median NRMSE by around 20 % compared to NTL, and it was particularly robust against unknown datasets. TimeMixer (42.2 %) and TiDE (45.3 %) performed particularly well in ZS mode.

The strong performance of ZS can likely be attributed to its extensive pre-training dataset comprising 108 heating load datasets containing 1.87 million hourly observations. This dataset exposes models to a variety of heating load-patterns across different building types, climate and operating conditions (Figs. A.13 and A.14). This considerable diversity of data enables the pre-trained models to capture generalisable temporal dependencies and load-patterns that can be effectively transferred to unknown buildings without the need for fine-tuning.

Notably, sequence-based models like PatchTST and TimeMixer, which rely exclusively on past heating load values without exogenous features, achieve some of the best ZS results, demonstrating that temporal patterns alone contain sufficient information for effective cross-building transfer. These results demonstrate the potential of ZS as a TL strategy in scenarios with limited data.

However, ZS can also exhibit characteristic failure modes. When the target building's load profile differs markedly from patterns seen during pre-training or contains abrupt load transitions (e.g. night setbacks), ZS forecasts may become over-smoothed and regress towards mean values. This behaviour is visible in the Serbia L8 dataset (Table 11), where errors increase particularly for MLP and TiDE, whereas PatchTST remain comparatively robust.

6.4. Why static features offer only limited advantages

This section clarifies that the empirical effect of SHAP-based SF is generally modest and often not statistically significant, despite the initial hypothesis of broader benefits. The results show that SF in the form of SHAP-based FI and load-pattern metrics bring no or only minimal improvements. In RQ1, it was assumed that such explicit SF could offer particular advantages in TL contexts by transferring building-specific insights between different datasets.

Lim et al. [23] demonstrated that, in the baseline scenario without TL, static covariates in TFT reduce the error by 2.6%. This formed the theoretical basis for our systematic investigation of SF in the TL context. We confirm this effect in the NTL case: TFT achieves an NRMSE of 0.316 without SF and 0.309 with SF, corresponding to an improvement of 2.2%, which closely matches the original findings. However, in TL scenarios—a context not investigated by Lim et al. [23]—SF effects are small and not consistently statistically significant across models and TL settings (FT, FTF, ZS). Under paired Wilcoxon tests, only isolated cases reach nominal significance (Table D.17). Across models, SF improves the forecast accuracy by 11.2% in NTL and 0.1% in FT. However, it deteriorates by −1.6% under FTF. An exception is TFT, which shows a 2.4% improvement under FT. Overall, the results show that SF offers advantages for individual models such as TFT. However, the improvement achieved through SF is modest and also depends on the TL strategy. For example, SF does not provide consistent gains in the ZS setting and often yields neutral or negative median changes across models in our experiments. TL likely reduces the need for explicit SF because transferable information is already contained in the temporal dynamics of the lagged hours (schedules, control behaviour, inertia). This is supported by our TFT attribution results, which demonstrate a strong shift towards autoregressive components. The internal TFT FI for past load increases from 0.218 under NTL, to 0.773 under FT and then to 0.913 under FTF. Essentially, the characteristic load profile of a building is already embedded in the time series, making additional static descriptors largely redundant for accurate transfer in our setting.

6.5. Practical implications

From a practical point of view, the advantages of sequence-based models, such as PatchTST or TimeMixer, outweigh the disadvantages in many applications. These models provide robust forecasts without the need for SF, require minimal memory and minimise the risk of negative transfers. Overall, it is evident that TL strategies, especially FTF and ZS, achieve high accuracy and are suitable for a broad range of applications in various buildings due to their robustness, even when data is limited. The results and findings presented thus contribute to the effective use of load forecasts for improved load flexibility.

6.6. Limitations

Despite taking a holistic approach wherever possible, this study has some limitations. The study does not include classical statistical or physics-informed baselines. Accordingly, the conclusions should be interpreted as comparisons among the evaluated neural architectures, datasets, and TL strategies rather than as a universal ranking of forecasting methods. For computational reasons, hyperparameter tuning was only performed on a single representative dataset, so the resulting settings may not be optimal for all datasets. In addition, when fine-tuning, the learning rate was reduced by a factor of 0.1, which was not systematically compared to alternative adaptation schedules. Similarly, the fine-tuning strategy with frozen layers relied on a single predefined layer-freezing configuration, without exploring alternative freezing schemes.

No systematic source selection of the datasets was performed. Instead, all the selected training data was used for pre-training to map a variety of different types of dataset. However, without source selection, negative transfers can also occur. Furthermore, using the training fraction rather than the number of training data leads to varying amounts of training data for the training fractions due to the different lengths of the datasets. Strict chronological 70/15/15 splits were employed to ensure a consistent evaluation protocol and to avoid look-ahead bias. Due to computational constraints, performance was not additionally validated using rolling-origin or time-block cross-validation.

The study focused on a specific set of SF derived from SHAP values and load-pattern metrics, and did not explore other XAI methods (e.g.

LIME, Permutation Feature Importance) or feature categories. Finally, the results may have been influenced by inaccuracies in the real-world measurements and external weather data. Furthermore, the value of the FIs which were used as SF, also depends on the time period for which the FI was determined, in this case the test period.

6.7. Future research

Future research should therefore systematically investigate the influence of different freezing layers and the adjustment of learning rates for the purpose of fine-tuning. Combining or using different XAI methods to identify robust SF, as well as comparing additional SF (e.g. building information or detailed usage profiles), can also be promising. Additionally, it would be beneficial to test TL approaches using different measurements or simulation data for various building types, usage types, and climate zones that have not yet been considered. This would increase the generalisability and practicality of the models. Although an unconstrained, data-driven design was chosen, hybrid physics-informed variants (e.g. PINNs) could enhance data efficiency and robustness to out-of-distribution data. Quantifying these trade-offs relative to the TL strategies would be a promising area for future research.

Examining how much data is required for an effective general heating load forecast is also useful, as is determining whether certain datasets in the training have reduced model accuracy. Since this study focused exclusively on a 24 h forecast horizon, future work should also evaluate the performance and transferability of models across longer forecast horizons.

7. Conclusion

Accurate forecasting of heating load is crucial for achieving good load flexibility in the heating sector, particularly given its increasing electrification. This study demonstrates that transfer learning—and, in particular, the selective freezing of network layers—can significantly enhance the quality of forecasts across various model architectures and datasets. Averaged across all models, fine-tuning with freezing achieved a 22% reduction in median NRMSE compared to models without transfer learning, with maximum improvements of up to 46.6% in specific cases such as TiDE. Even when using just 10% of the training data, the forecast accuracy remained stable for models such as PatchTST and TiDE when fine-tuning. Zero-shot inference, which does not require area-specific training data for the target region, achieved a comparable level of performance, reducing error by 20% compared to NTL, and exhibited the highest robustness for previously unknown datasets. This makes ZS a particularly promising strategy for data-poor application scenarios.

Including additional static features, such as SHAP-based feature importances or load-pattern metrics, led to only minor or inconsistent improvements on average. Small advantages were mainly observed in individual TFT and MLP fine-tuning configurations, while other architectures showed negligible or even slightly negative effects in some TL settings. Forecasting models such as PatchTST and TiDE proved to be exceptionally powerful and resource-efficient: PatchTST achieved a median NRMSE of 0.315 under FTF with minimal memory requirements and without exogenous features.

Overall, the results show that targeted transfer learning approaches—especially FTF and ZS—can effectively mitigate the problem of limited data availability in heating load forecasting. These accurate forecasts are essential for energy management systems and load-shifting strategies to optimise demand-side flexibility in the building sector. Future research should explore transferability to different building types and climate conditions, adaptive strategies for fine-tuning, freezing, and learning rates.

Table A.13

Overview of the datasets with training and test periods, dataset source, and number of samples. This summary provides context on the temporal coverage and allocation of each dataset to training or evaluation (seen/unseen) scenarios.

Train period	Dataset source	Num. Train data	Num. Test data	Start train	End train	Start test	End test
Long	Denmark DHS	18 270	3915	2018-01-08	2020-02-08	2020-07-20	2020-12-30
	Serbia DHS	7508	1609	2023-01-09	2023-11-17	2024-01-24	2024-03-31
	Germany GF	9450	2025	2023-10-15	2024-11-12	2025-02-04	2025-04-29
	Germany Hotel	17 593	3770	2022-01-06	2024-01-09	2024-06-14	2024-11-18
	Germany VDI	6024	1291	2018-01-06	2018-09-14	2018-11-07	2018-12-31
	BDG2	12 054	2583	2016-01-12	2017-05-28	2017-09-13	2017-12-30
Short	Denmark DHS	1827	3915	2019-11-24	2020-02-08	2020-07-20	2020-12-30
	Serbia DHS	750	1609	2023-10-17	2023-11-17	2024-01-24	2024-03-31
	Germany GF	945	2025	2024-10-03	2024-11-12	2025-02-04	2025-04-29
	Germany Hotel	1759	3770	2023-10-28	2024-01-09	2024-06-14	2024-11-18
	Germany VDI	602	1291	2018-08-20	2018-09-14	2018-11-07	2018-12-31
	BDG2	1205	2583	2017-04-08	2017-05-28	2017-09-13	2017-12-30

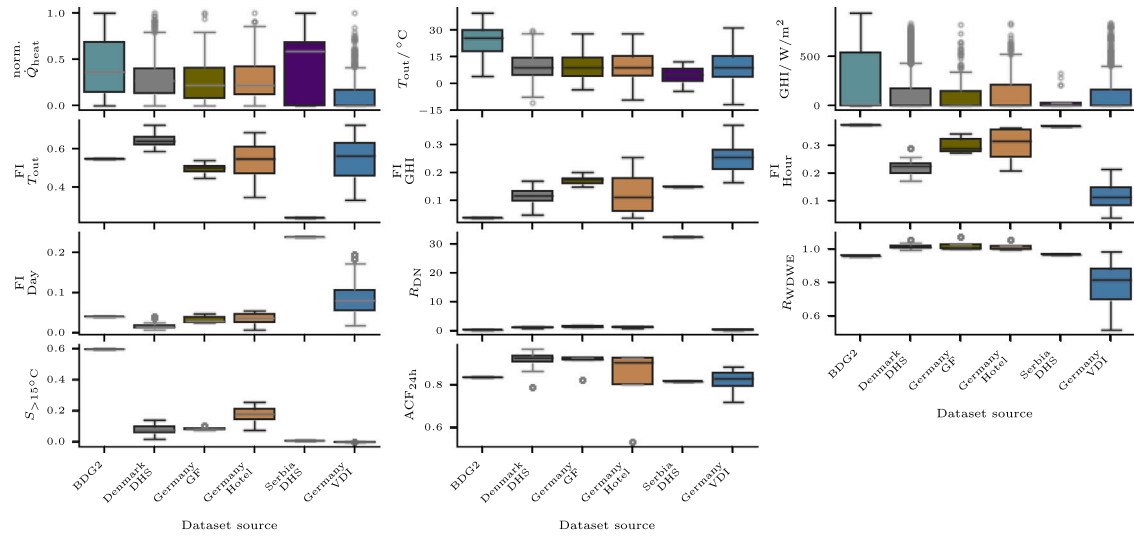


Fig. A.13. Distributions of dynamic (past/future) and static features, as well as the label (heating load), are shown across different dataset sources. This includes variability in weather conditions and SHAP-based feature importances. Such comparisons highlight differences in building usage, climate and feature relevance that could impact model training and transferability.

CRedit authorship contribution statement

Alexander Neubauer: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Mengbo Yu:** Writing – review & editing. **Pedram Babakhani:** Writing – review & editing. **Stefan Brandt:** Writing – review & editing, Project administration, Funding acquisition. **Martin Kriegel:** Supervision, Resources.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used DeepL and Perplexity in order to improve the readability and language of the manuscript. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Details about the datasets

This section provides information about the datasets used. Table A.14 shows a summary of all the datasets used for pre-training and the subsequent evaluation. This includes the dataset IDs, the types of building used, the time period for training and test and the assignment to the training, test (seen) and test (unseen) sets.

Fig. A.13 shows the distributions of all the static and dynamic features used, as well as the normalised label ($\dot{Q}_{\text{heat}}$), grouped by dataset sources and in the form of box plots. It is therefore possible to observe differences in the distribution of $\dot{Q}_{\text{heat}}$, weather conditions and SHAP-based FI between different building types and data sources. In many datasets, T_{out} drop below -10°C , and sometimes even to -15°C , which reflects the climate zones of most cities. In the FI, the T_{out} is usually the highest. For the R_{DN} and R_{WDWE} , you can see the differences for

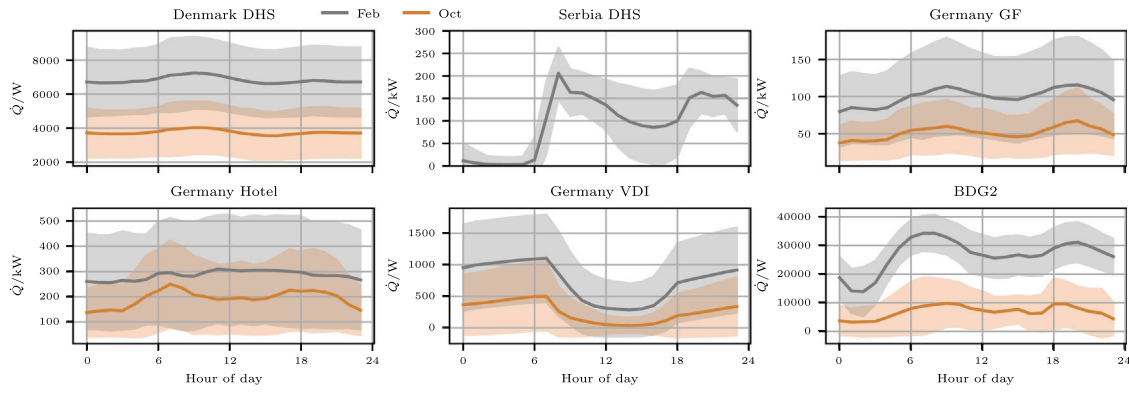


Fig. A.14. Mean load profiles including standard deviation range of the various dataset sources for February (Feb) and October (Oct).

Table A.14

Overview of the datasets used for pre-training and testing (both seen and unseen).

Dataset source	Only pre-train set	Pre-train+test (seen)	Only test (unseen)
Denmark DHS	45, 59, 222, 317, 432, 454, 490, 613, 672, 746, 849, 853, 862, 872, 888, 900, 998, 1241, 1337, 1413, 1414, 1572, 1625, 1900, 1990, 2117, 2145, 2165, 2196, 2323, 2382, 2388, 2410, 2468, 2531, 2537, 2550, 2563, 2609, 2652, 2760, 2791, 2806, 2912, 2982, 3002, 3033, 3088, 3124	1900	3002, 672
Serbia DHS	L22	L22	L12, L8
Germany GF	229, 289, 290, 292, 295, 407	229, 289, 290, 407	401, 409
Germany Hotel	2, 3, 4, 6	2, 3	5
Germany VDI	Berlin room1 weight-l rt-3 g-0.3 i-0.75 a-0.75, Berlin room1 weight-l rt-5 g-0.3 i-0.75 a-4, Berlin room1 weight-l rt-5 g-0.9 i-0.25 a-0.5, Berlin room1 weight-l rt-5 g-0.9 i-1 a-4, Berlin room1 weight-m rt-5 g-0.45 i-0.75 a-1, Berlin room1 weight-m rt-5 g-0.6 i-1.5 a-2, Berlin room2 weight-l rt-1 g-0.45 i-0.5 a-4, Berlin room2 weight-l rt-1 g-0.75 i-0.5 a-0.5, Berlin room2 weight-l rt-1 g-0.75 i-1 a-4, Berlin room2 weight-l rt-2 g-0.15 i-0.5 a-0.5, Berlin room2 weight-l rt-2 g-0.15 i-1.25 a-1, Berlin room2 weight-l rt-3 g-0.6 i-1.25 a-1, Berlin room2 weight-l rt-4 g-0.75 i-1 a-1, Berlin room2 weight-l rt-5 g-0.45 i-1.25 a-1, Berlin room2 weight-l rt-5 g-0.6 i-0.25 a-0.5, Berlin room2 weight-l rt-5 g-0.75 i-0.75 a-0.75, Berlin room2 weight-l rt-5 g-0.75 i-1.5 a-0.75, Berlin room3 weight-l rt-2 g-0.15 i-1.25 a-1, Berlin room3 weight-l rt-3 g-0.15 i-1 a-2, Berlin room3 weight-l rt-5 g-0.9 i-1 a-4, Berlin room3 weight-m rt-2 g-0.3 i-1 a-0.25, Berlin room3 weight-m rt-3 g-0.45 i-0.75 a-1, Berlin room3 weight-m rt-3 g-0.75 i-1 a-4, Berlin room3 weight-m rt-4 g-0.6 i-1.5 a-2, Berlin room3 weight-m rt-4 g-0.6 i-0.75 a-1, Berlin room3 weight-m rt-4 g-0.9 i-0.5 a-0.25, Berlin room3 weight-m rt-5 g-0.3 i-0.5 a-1, Stuttgart room1 weight-l rt-4 g-0.3 i-0.5 a-0.75, Stuttgart room1 weight-l rt-4 g-0.75 i-0.5 a-4, Stuttgart room2 weight-l rt-4 g-0.45 i-0.25 a-4, Stuttgart room2 weight-l rt-5 g-0.45 i-0.5 a-0.25, Stuttgart room2 weight-l rt-5 g-0.45 i-1 a-2, Stuttgart room2 weight-m rt-2 g-0.3 i-0.5 a-0.75, Stuttgart room2 weight-m rt-3 g-0.45 i-0.25 a-0.5, Stuttgart room2 weight-m rt-3 g-0.45 i-1 a-1, Stuttgart room2 weight-m rt-4 g-0.3 i-0.25 a-0.75, Stuttgart room2 weight-m rt-4 g-0.6 i-0.25 a-0.5, Stuttgart room2 weight-m rt-4 g-0.6 i-0.5 a-2, Stuttgart room2 weight-m rt-5 g-0.6 i-0.5 a-2, Stuttgart room3 weight-l rt-2 g-0.6 i-0.5 a-0.75, Stuttgart room3 weight-l rt-3 g-0.45 i-0.75 a-0.75, Stuttgart room3 weight-l rt-4 g-0.15 i-0.5 a-2, Stuttgart room3 weight-m rt-4 g-0.45 i-1.25 a-0.75, Stuttgart room3 weight-m rt-4 g-0.9 i-0.5 a-0.25, Stuttgart room3 weight-m rt-4 g-0.9 i-0.75 a-0.25, Stuttgart room3 weight-m rt-5 g-0.15 i-1.25 a-2	Stuttgart room3 weight-m rt-4 g-0.9 i-0.75 a-0.25	Stuttgart room2 weight-m rt-3 g-0.45 i-0.25 a-0.5
BDG2	Fox	Fox	—

the Serbia DHS, which sometimes experiences a significant reduction in $\dot{Q}_{\text{heat}}$ at certain times, and the Germany VDI data source, which is an office building.

Appendix B. Static feature equations

The following SF were calculated for each dataset and used as additional inputs to the models. They quantify characteristic heating load-patterns and their relationship with the outdoor temperature. All SF were scaled using a robust scaler fitted on the training set of the global dataset.

1. Day-Night Load Ratio (R_{DN})

Captures differences in average temperature-normalised load between daytime (8:00–18:00) and nighttime (0:00–4:00) hours during the heating period, indicating night setbacks or typical occupancy-related load-patterns.

$$R_{\text{DN}} = \frac{\frac{\bar{\dot{Q}}_{\text{day}}}{15 - \bar{T}_{\text{day}}}}{\frac{\bar{\dot{Q}}_{\text{night}}}{15 - \bar{T}_{\text{night}}}} \quad (\text{B.1})$$

Table C.15

Hyperparameter comparison across seven models under no transfer learning (NTL) and transfer learning with fine-tuning (FT). Model mapping: M1 = MLP, M2 = TiDE, M3 = TimeMixer, M4 = LSTM, M5 = TFT, M6 = PatchTST, M7 = Time-LLM.

Hyper-parameter	Models used in	Search Range/Values	Type	No transfer learning (NTL)							Fine-tune (FT)						
				M1	M2	M3	M4	M5	M6	M7	M1	M2	M3	M4	M5	M6	M7
Input size	all	[24, 48, 72, 96]	choice	48	96	96	24	24	96	–	24	24	48	72	96	24	–
Learning rate	all	[1e-4, 1e-2]	log-uniform	0.000185	0.002397	0.000143	0.000110	0.002274	0.001524	–	0.004484	0.001995	0.000124	0.004337	0.001904	0.000214	–
Max training steps	all	[300–10000]	choice	300	300	2000	2000	2000	300	–	1000	5000	10000	10000	10000	10000	–
Batch size	all	[16, 32, 64]	choice	64	32	16	64	16	32	–	32	32	32	32	64	32	–
Windows batch size	all	[128, 256, 512, 1024]	choice	512	1024	128	1024	128	256	–	512	128	256	512	512	128	–
Scaler type	all	robust (fixed)	fixed	robust	robust	robust	robust	robust	robust	–	robust	robust	robust	robust	robust	robust	–
Random seed	all	[1–100]	randint	40	56	9	36	61	88	–	72	36	89	55	37	21	–
Start padding enabled	all	[True, False]	fixed	True	True	True	True	True	True	–	True	True	True	True	True	True	–
Dropout rate	M2–M6	[0.05, 0.3]	uniform	–	0.147	0.239	0.278	0.265	0.135	–	–	0.276	0.287	0.224	0.255	0.271	–
Hidden size	TiDE, TimeMixer, LSTM, TFT, PatchTST	[16, 32, 64, 128, 256]	choice	–	64	–	64	32	64	–	–	32	–	256	32	32	–
Patch length	PatchTST	[12, 24]	choice	–	–	–	–	–	24	–	–	–	–	–	–	12	–
Stride	PatchTST	[1, 12, 24]	choice	–	–	–	–	–	24	–	–	–	–	–	–	24	–
Number of heads	TFT	[2, 4, 8]	choice	–	–	–	–	4	–	–	–	–	–	–	2	–	–
Attention dropout	TFT	[0.05, 0.3]	uniform	–	–	–	–	0.067	–	–	–	–	–	–	0.202	–	–
GRN activation	TFT	ReLU, GELU, etc.	choice	–	–	–	–	GELU	–	–	–	–	–	–	ReLU	–	–
Number of RNN layers	TFT	[1, 2, 3]	choice	–	–	–	–	3	–	–	–	–	–	–	2	–	–
RNN type	TFT	[LSTM, GRU]	choice	–	–	–	–	LSTM	–	–	–	–	–	–	LSTM	–	–
Encoder hidden size	LSTM	[32, 64, 128, 256]	choice	–	–	–	64	–	–	–	–	–	–	256	–	–	–
Decoder hidden size	LSTM	[32, 64, 128, 256]	choice	–	–	–	32	–	–	–	–	–	–	256	–	–	–
Encoder dropout	LSTM	[0.05, 0.3]	uniform	–	–	–	0.278	–	–	–	–	–	–	0.224	–	–	–

2. Weekday–Weekend Load Ratio (R_{WDWE})

Compares temperature-normalised mean loads on weekdays versus weekends, highlighting usage profiles such as reduced operation during weekends for office-type buildings.

$$R_{WDWE} = \frac{\frac{\bar{Q}_{\text{weekday}}}{15 - \bar{T}_{\text{weekday}}}}{\frac{\bar{Q}_{\text{weekend}}}{15 - \bar{T}_{\text{weekend}}}} \quad (\text{B.2})$$

3. Autocorrelation at 24 h lag (ACF_{24h})

Measures daily periodicity by computing the autocorrelation of the hourly heating load at a lag of 24 h, indicating how strongly consecutive days follow similar load-patterns.

$$ACF_{24h} = \rho(24) \quad (\text{B.3})$$

4. Share of Load above 15 °C ($S_{>15^\circ\text{C}}$)

Fraction of total heating energy consumed when outdoor temperatures exceed 15 °C, indicating possible contributions from domestic hot water, ventilation, or process loads.

$$S_{>15^\circ\text{C}} = \frac{\sum_{T_{\text{out}} > 15^\circ\text{C}} \dot{Q}}{\sum \dot{Q}} \quad (\text{B.4})$$

Appendix C. Hyperparameter tuning and frozen layers

Table C.15 lists the hyperparameters used in this study, along with their respective search ranges and optimal values. The hyperparameter tuning for the NTL and NTL+SF variants was conducted on Denmark DHS dataset 490, whereas for FT and FT+SF, it was performed on the entire pre-training set, which comprises all pre-training datasets. In addition to general parameters such as batch size, maximum training steps, and lag hours, architecture-specific hyperparameters were also optimised. For early stopping, both settings used a validation check every 50 steps, with a patience of 15 steps for NTL and 10 steps for FT. This overview ensures transparent traceability of the selected model configurations and provides a basis for future comparative studies.

Table C.16

Frozen layers for selective fine-tuning strategy.

Model	Typical layers/blocks frozen
MLP	Initial linear layer(s)
TiDE	Dense encoder component
TimeMixer	Input patching/embedding layers
LSTM	Historical encoder part
TFT	Temporal encoder, Temporal fusion decoder components
PatchTST	Input patching and embedding layer
Time-LLM	Core embedding layers and initial attention blocks

Fig. C.15 (NTL) and Fig. C.16 (FT) show the model validation MSE as a function of different hyperparameters. The y-values are limited to the 95th percentile. The categories for the learning and dropout rates were defined using fixed interval limits, with the x-values indicating the respective minimum values of the individual categories.

In the NTL case, PatchTST is highly sensitive to different hyperparameter values, whereas in the FT scenario, TFT in particular is highly sensitive to various settings. For instance, in the NTL case, PatchTST only achieves low validation losses with a step size of 96, whereas in the pre-train case, low validation losses are achieved with even small input sizes. Therefore, careful hyperparameter tuning is essential for the different models and variants, as the optimal hyperparameters for NTL and pre-train can differ significantly.

Table C.16 outlines the specific layers or blocks that are frozen during the selective layer freezing fine-tuning strategy for each of the evaluated model architectures. The layer names correspond to their typical implementation in common forecasting libraries or their conceptual role. Exact parameter names may vary slightly based on the specific library version used. The description of the selection of the frozen layer can be found in Section 4.

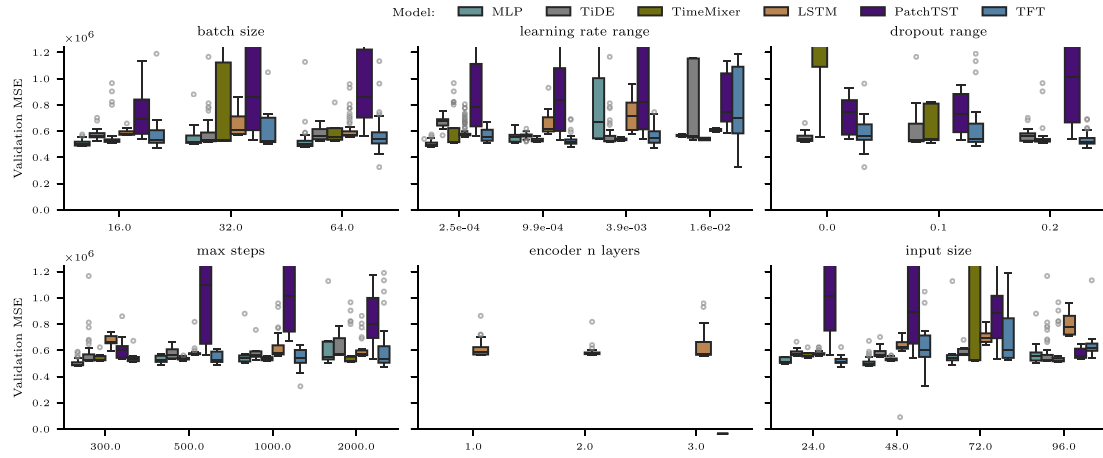


Fig. C.15. Influence of various hyperparameters on validation MSE in hyperparameter tuning for the NTL strategy.

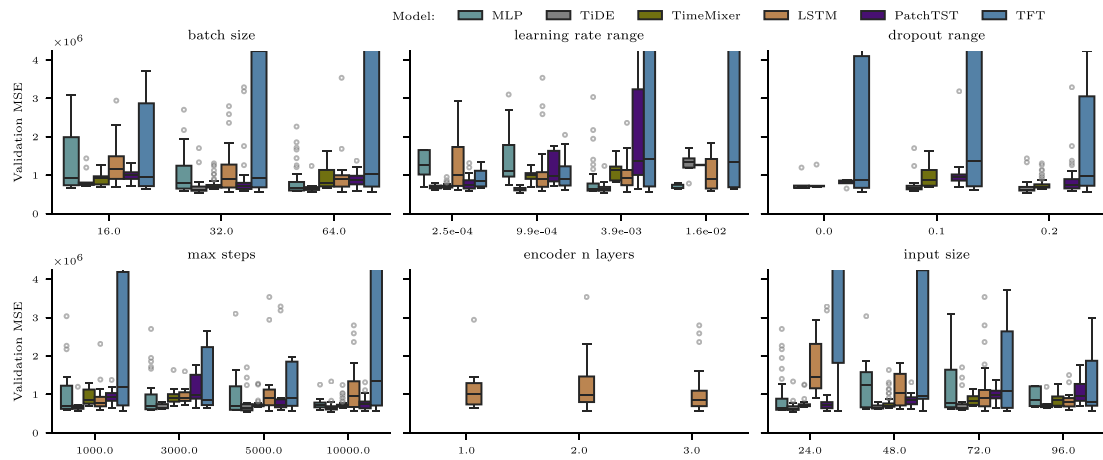


Fig. C.16. Influence of various hyperparameters on validation MSE in hyperparameter tuning for the FT TL strategy.

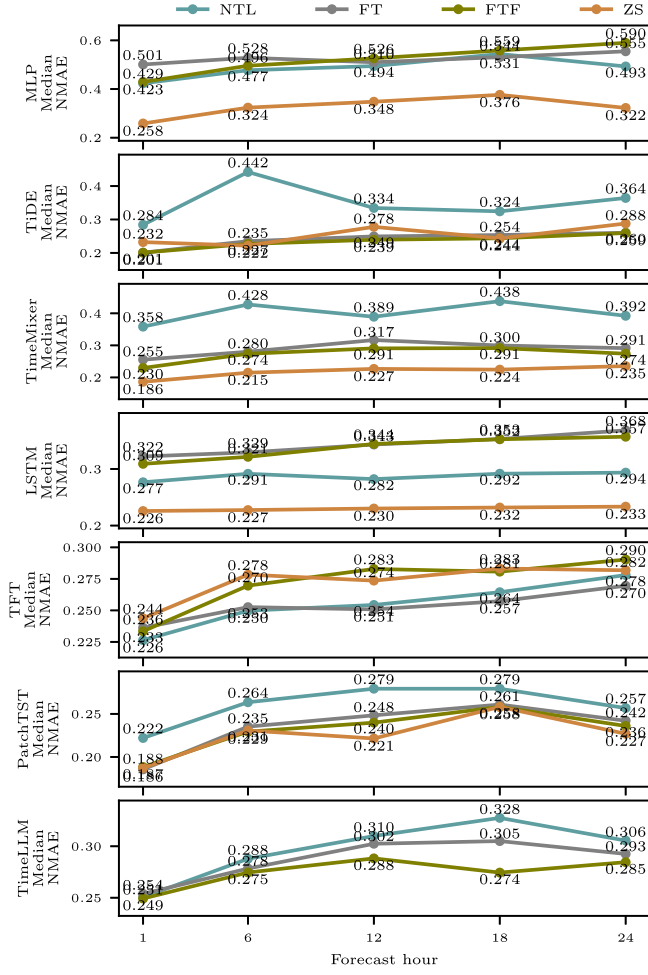


Fig. D.17. NMAE of the different models and TL strategies depending on the forecast hour.

Table D.17

Significance test results (p -values) for static feature impact and transfer learning effects, both evaluated with the Wilcoxon signed-rank test on paired datasets. Bold values indicate statistically significant differences ($p < 0.05$). Static features were not supported (and therefore not used) for PatchTST, TimeLLM, and TimeMixer (Table 2).

Test		LSTM	MLP	TFT	TiDE	Patch TST	Time LLM	Time mixer
Static vs. no Static	NTL	0.2121	0.5798	0.2462	0.7660	–	–	–
	FT	0.0483	0.0814	0.0987	0.3173	–	–	–
	FTF	0.2842	0.8100	1.0000	0.3431	–	–	–
	ZS	0.9661	0.1430	0.1187	0.5402	–	–	–
TL vs. NTL	FT vs.NTL	0.0120	0.7660	0.9661	0.0003	1.0000	0.0077	0.0056
	FTF vs.NTL	0.0665	0.4951	0.1815	0.0003	0.6777	–	0.0129
	ZS vs.NTL	0.0539	0.0003	0.0066	0.6705	0.9323	0.0833	0.1815

Appendix D. Supplementary results to the main text

Figs. D.17 and D.18 show the median NMAE and NRMSE, respectively, for the different models and TL strategies depending on the forecast hour. For most models, forecast accuracy decreases with increasing forecast hour. However, there are some exceptions, such as ZS for LSTM, where the forecast accuracy remains almost constant.

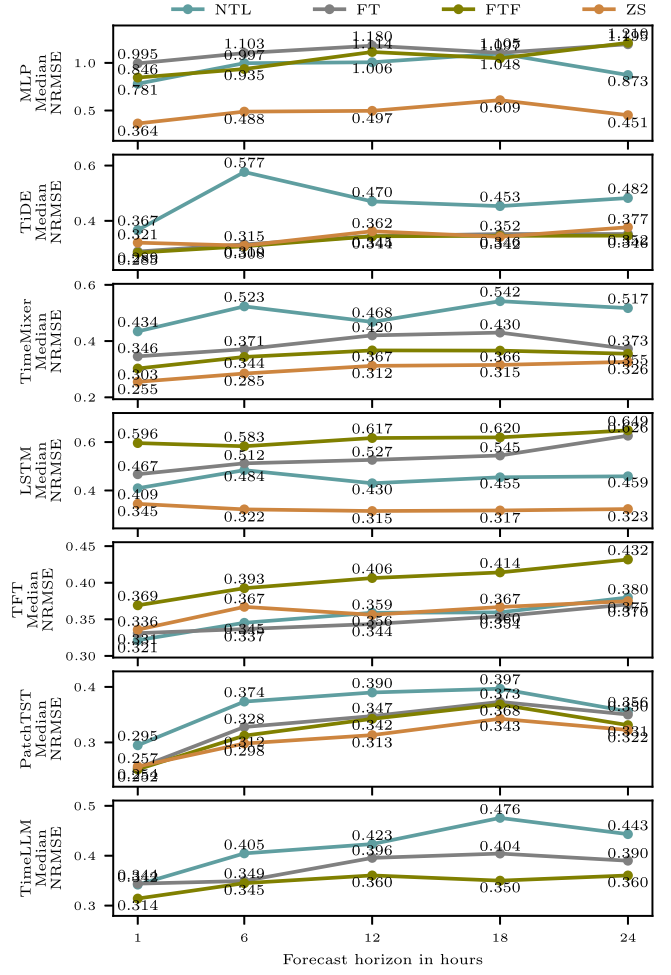


Fig. D.18. NRMSE of the different models and TL strategies depending on the forecast hour.

Table D.18

Mean absolute FI values of the TFT model for the NTL, FT, and FTF variants. The values are grouped by feature category and sorted by FI.

Category	Feature	NTL	FT	FTF
Past	T_{out}	0.203	0.049	0.024
	GHI	0.220	0.042	0.014
	Hour	0.195	0.033	0.026
	Day	0.164	0.103	0.023
	Q_{heat}	0.218	0.773	0.913
Future	T_{out}	0.255	0.306	0.309
	GHI	0.216	0.356	0.356
	Hour	0.280	0.288	0.290
	Day	0.250	0.050	0.046
Static	$FI_{T_{out}}$	0.056	–	–
	FI_{GHI}	0.095	–	–
	FI_{Hour}	0.185	–	–
	FI_{Day}	0.042	–	–
	R_{DN}	0.202	–	–
	R_{WDWE}	0.104	–	–
	$ACF_{24\ h}$	0.227	–	–
	$S_{>15^{\circ}C}$	0.088	–	–

Table D.19

Generalisation gap (4_{err} in NRMSE $\times 100$) between seen and unseen datasets for each model and transfer strategy. Smaller values indicate better generalisation.

Model	FT vs NTL	FTF vs NTL	ZS vs NTL
MLP	-12.33	10.72	18.06
TiDE	6.32	12.82	13.17
TimeMixer	7.00	9.17	10.65
LSTM	8.15	4.76	9.72
TFT	-0.01	3.34	15.01
PatchTST	9.93	13.46	10.25
Time-LLM	12.82	-	20.77

Data availability

Four of six datasets are publicly available with cited sources; one dataset is restricted, others can be shared upon reasonable request.

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